

Chapter 4

Aqueous Cleaning and Surface Conditioning Processes

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4.1 OVERVIEW OF AQUEOUS CLEANING, RINSING, AND DRYING APPLICATIONS AND TECHNIQUES

Fabrication of integrated circuit (IC) devices requires contamination removal, particle removal, and surface conditioning to achieve high-yielding products. Most cleaning and surface conditioning processes are performed in the liquid phase, and a majority of these processes are aqueous. Liquid cleaning offers several advantages, including high solubility of chemical reagents and contaminants, drag forces to aid in removal of solid contaminants, metal complexation, and easier avoidance of electrostatic deposition of particles compared with air. Aqueous liquids are further advantageous, in part because water is easily and inexpensively purified so that ultrapure deionized water (DIW) is abundantly available in semiconductor factories. Thus by using aqueous cleaning processes, the requirements of achieving extremely high chemical purity, maintaining low manufacturing cost, and minimizing the

adverse environment and health effects, which are constant themes for semiconductor manufacturing engineers, are more manageable.

Wet cleaning processes can be categorized as preprocess cleaning or postprocess cleaning steps in the IC manufacturing flow. Furthermore, they can be divided between the front-end-of-line (FEOL), middle-of-line (MOL), and back-end-of-line (BEOL), FEOL and MOL essentially comprising steps to form the active devices and the BEOL steps to connect them by multiple levels of metal wiring.

Mixing of H₂O with various chemicals forms the most common cleaning solutions used in IC manufacturing, especially for FEOL and MOL. Typically, these chemicals are mixed with DIW at the manufacturing facility. [Section 4.2](#) discusses FEOL stripping, cleaning, and surface conditioning processes.

Historically, BEOL cleaning steps have been accomplished using organic solvents as opposed to aqueous chemistries because of the incompatibility of FEOL aqueous cleaning solutions with the metal wiring. The BEOL has numerous specialty chemicals, which are delivered preblended from chemical suppliers. These chemicals may be organic solvent-based or aqueous-based. In recent years, aqueous cleaning has become increasingly used in the BEOL. [Section 4.3](#) discusses BEOL stripping and cleaning processes.

The chemicals used for cleaning will be discussed in [Section 4.4](#), whereas the variables affecting the cleaning process will be discussed in [Section 4.5](#). The cleaning process must be followed by rinsing and drying of the wafer, discussed in [Section 4.6](#). Equipment for the cleaning and stripping processes will be presented in [Section 4.7](#), including batch immersion and spin-spray, both for single wafers and multiple wafer batches. The chapter will wrap up with future challenges in [Section 4.8](#).

4.2 FEOL CLEANING

4.2.1 FEOL Preprocessing Cleaning

The general term of preprocess cleaning, also known as precleaning, refers to cleaning steps immediately before thermal processing, such as oxidation, deposition, and annealing, and sometimes lithography. In such circumstances, it is essential to have a surface that is free from particulate, metallic, and organic contaminants before the next step. In some cases, such as deposition of epitaxial silicon (Si), it may also be necessary to have a hydrogen-terminated (H-terminated) Si surface that is free from native or chemical oxide. The terms “surface preparation” or “surface conditioning” are often used in place of “cleaning,” since the goal is in fact to chemically prepare a surface for the subsequent process.

Although the original formulation has been modified over the years and alternatives have been proposed (discussed in [Section 4.4.4](#)), the RCA cleaning sequence developed in the 1960s, by Werner Kern et al. [1–3], has

proved remarkably durable and effective, and it remains the backbone of most preprocess cleaning in the IC manufacturing industry today. In the two-step immersion cleaning sequence, wafers are contacted by a first solution called SC-1 and a second solution called SC-2. The SC-1 is a dilute aqueous mixture of ammonium hydroxide (NH_4OH) and hydrogen peroxide (H_2O_2), an oxidizer; hence the solution is sometimes referred to as APM (ammonium hydroxide and hydrogen peroxide mixture). SC-1 is used primarily to remove particles, although it is also capable of removing light organic contamination and complexing some metals such as Cu and Zn. Other metals such as Fe, however, are easily deposited onto the silicon surface from SC-1 solutions. For effective removal of alkaline and a wide range of metal contaminants, the SC-2 aqueous solution comprising hydrochloric acid (HCl) and H_2O_2 is used; the solution is sometimes referred to as HPM (hydrochloric acid and hydrogen peroxide mixture). The RCA cleaning steps form an integral part of the commonly used precleaning sequences. The overall strategy of such a sequence is typically as follows, with intermediate DIW rinsing steps separating each chemical step:

1. Removal of relatively heavy organic contamination, such as residual photoresist.
2. Removal of silicon oxide, either in bulk or the very thin (chemical) oxide formed by the organic removal step.
3. Particle and metal removal by SC-1, with chemical oxide regrowth.
4. Residual metal removal by SC-2 (including metals that may have been deposited in the SC-1 solution).
5. Final rinsing and drying.

A detailed description of the original RCA cleaning process and its subsequent modifications is presented in Chapter 1.

Table 4.2-1 shows typical chemicals used for the precleaning process. The sulfuric acid (H_2SO_4) and H_2O_2 mixture (SPM is sulfuric acid/hydrogen peroxide mixture) also is relied upon for the postcleaning process. Hydrofluoric acid (HF) is widely used for silicon dioxide (SiO_2) removal. The chemical composition and properties of the solutions are shown in Table 4.2-2. The sequences of the cleaning steps are shown in Table 4.2-3.

4.2.2 FEOL Resist Stripping and Postprocess Cleaning

Postprocess cleaning involves removal of all the contaminants previously mentioned—particulate, metallic, and thin organic layers plus others, such as residual slurry, thick organic films and polymers, and film passivation layers. Three common types of postcleaning processes occur after resist stripping, also known as ashing, after reactive ion etching (RIE), and after chemical mechanical planarization (CMP). Common postcleaning sequences are shown in Table 4.2-4.

TABLE 4.2-1 RCA Chemistries Showing Typical Dilution and Temperature Ranges Used for IC Manufacturing

RCA Solution	Chemicals and Conditions	Contaminant Removal
SPM	H ₂ SO ₄ and H ₂ O ₂ , 2:1–4:1, 90–140°C	Organics
SC-1	NH ₄ OH:H ₂ O ₂ :H ₂ O, 1:1:5–1:1:100, 40–75°C	Organics, some metals, particles, surface conditioning of Si along with DHF (dilute HF)
SC-2	HCl:H ₂ O ₂ :H ₂ O, 1:1:6–1:1:50, 40–75°C	Metals: alkaline and trace metals
DHF	HF, 1:10–1:200, 25°C	Sacrificial oxide removal, surface conditioning, native oxide removal

TABLE 4.2-2 Chemicals Used for Cleaning and Drying Processes and Some of Their Properties

Chemical Formula	Chemical Name	Typical Commercially Available Concentration (wt%)	Surface Tension at 20°C (dynes/cm)
H ₂ O	Water	N/A	73
NH ₄ OH	Ammonium hydroxide	25.5–29	63
HCl	Hydrochloric acid	15–37	65
HF	Hydrofluoric acid	49	55
NH ₄ F:HF (BHF)	Ammonium fluoride: Hydrofluoric acid	10:1 volume ratio 49	80–90
H ₂ SO ₄	Sulfuric acid	95–100	75
H ₂ O ₂	Hydrogen peroxide	30–40	74–78
CH ₃ –CH(OH)–CH ₃	Isopropyl alcohol (IPA)	100	21.7

TABLE 4.2-3 Pre-thermal Processing Cleaning Sequences and Chemistries	
Cleaning Step	Typical Sequences
Preoxidation (prepad or noncritical oxidation)	SPM—DHF—SC-1—SC-2
Preoxidation (pregate or other critical oxidation)	SC-1—SC-2—DHF
Preanneal/prespacer	SC-1—SC-2
Precontact/presilicide	SC-1—DHF or BHF
Precontact/presilicide	DHF

TABLE 4.2-4 Postprocessing Cleaning Sequences and Chemistries	
Cleaning Steps	Typical Sequence
FEOL poststrip clean or photoresist strip and clean	SPM—SC-1
FEOL postetch clean	DHF—SPM—SC-1
<i>FEOL</i> , front-end-of-line; <i>SPM</i> , sulfuric acid/hydrogen peroxide mixture.	

4.2.2.1 FEOL Photoresist Removal

Photoresist, sometimes simply called resist, is used as a mask for the formation of patterned etched features or for ion implantation. Stripping the organic material requires postetch or postimplant processing by dry and/or wet methods. Most commonly a combination of these methods is used. A plasma process is typically used to strip the bulk photoresist, followed by a wet cleaning process such as hot SPM (typically >120°C) to remove any remaining residues. Methods using ozone (O₃) gas, either as a replacement for H₂O₂ in SPM (known as SOM) or simply dissolved in H₂O (DIW/O₃), have been developed as alternatives to SPM. Photoresist is an organic material containing long-chain polymer, so oxidation is critical to its breakdown and removal. The difficulty of resist removal depends on prior processing. These processes can include: (1) baking the resist, which hardens it by degassing; (2) exposure of the photoresist in the photolithography tool, which cross-links the portion of resist that is not developed; and (3) ion implantation at a high dose, which heavily cross-links and dehydrogenates the polymer into a hard “crust” near the outer surface of the resist [4]. Historically, aqueous-based chemistry methods have been capable of removing photoresist that has not been ion implanted or that has been implanted at a dose less than about 1.0 × 10¹⁴ atoms/cm²; however, higher dose implanted resists have required plasma stripping before wet cleaning. In recent years, however, more

aggressive SPM-based processes have proved effective at removing implanted resists across a wide range of doses without the need for plasma stripping.

4.2.2.2 FEOL Postetch Cleaning

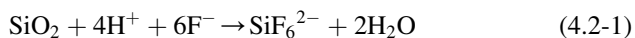
One advantage of the RIE process, for pattern definition using a photoresist mask, is the anisotropic nature of the plasma resulting in vertical profiles without horizontal etching. This enables the formation of trenches, contact holes, and vias. The anisotropic etch is achieved by a combination of ion directionality and the formation of solid residue, the so-called etch polymer, during the RIE process. The residue, however, must be subsequently removed, a typical task for wet cleaning processes, frequently in conjunction with plasma stripping of the photoresist mask. The residue contains both organic and inorganic (often Si-containing and sometimes metal-containing) materials that must be removed. After etching of the gate stack polysilicon (polySi), for example, a sequence including SPM and HF is normally used for removal of both etch and strip residues, in addition to unwanted gate oxide that remains in the etched areas.

4.2.3 FEOL Oxide Etching and Removal

An important part of aqueous cleaning is the removal of oxide layers that are present before the cleaning process, or created by the cleaning steps themselves. These oxides may contain contaminants or may be sacrificial (temporarily used) oxides that subsequently must be removed. This is accomplished using hydrofluoric acid (HF), either diluted with H₂O (DHF) or pH buffered using ammonium fluoride (NH₄F), known as buffered HF (BHF) or buffered oxide etch (BOE). These chemistries will be discussed in detail in [Section 4.4.2.2](#).

A critical difference between dry etching using RIE and wet etching is that the latter is isotropic. Etching will occur in all directions, which makes wet etching effective for film removal or etch-back but inferior to RIE processes for unidirectional etch requirements, such as trenches and vias. Also, the isotropic nature of wet etching makes undercut (undesired lateral etching) a concern in some cases.

Concentrated HF is actually a 49 wt% solution of HF in H₂O, at a density of 1.15 g/mL, thus a volume dilution of 10:1 HF indicates ~5.1 wt% HF, 100:1 indicates ~0.56 wt%, and so on. Dilutions ranging from 5:1 to 200:1 commonly are used in the industry, and in recent years higher dilutions are sometimes used. A hydrofluoric acid solution reacts with SiO₂ as follows [5]:



The etching rate increases with temperature and concentration. Typically, room temperature is used, whereas time and concentration determine the amount of film removal.

In BHF solutions, 40 wt% NH_4F is added to the aqueous HF solution. The NH_4F dissociates to provide F^- ions that stabilize etching rates by replacing ions consumed in the SiO_2 etch reaction, hence the buffered solution. Such solutions may also contain surfactants that aid in the wetting of high aspect ratio trench and contact structures and in preventing Si attack for higher pH formulations. Buffered HF or DHF solutions are often used with photoresist formulations that may undesirably react with concentrated HF solutions.

4.3 BEOL CLEANING

The most common application for BEOL wet cleaning is after dry etching processes. Because of the increasing number of interconnect layers and the decreasing feature sizes there is a constant effort to develop effective postetch residue (PER) removal processes while minimizing wafer contamination. The BEOL etch-generated residues are not well-defined compounds but mixtures of metal, metal oxides, Si, oxygen, carbon, and etching gas residues that sometimes contain fluorine (F) or chlorine (Cl) atoms. Besides removing PER and possibly the photoresist used to pattern the device features, it is also critical to remove surface contaminants such as mobile ions and particles.

The cleaning applications for BEOL include Al interconnect structures, via structures landing on metal, both Al and Cu, and trench structures for dual Damascene processes. Materials such as Cu (copper) and low-k (dielectric constant) dielectrics, along with small linewidths and high aspect ratios for both Al (aluminum) and Cu feature, require unique cleaning solutions developed specifically for BEOL processing. Depending on the device type (logic, memory, etc.), the types of material (Al or Cu, low-k), the BEOL interconnect scheme, and the number of interconnect layers, there could be as many as 50 BEOL cleaning steps.

4.3.1 Aluminum Interconnect Cleaning

The Al metallization process involves either the subtractive etching of metal to form the metal interconnect lines or etching vias through silica to a previous metal layer. Each etching process generates a different type of PER. After etching the Al metal interconnect, cleaning typically consists of an in situ downstream O_2 or H_2/O_2 -based plasma step for resist removal, followed by wet chemistry. Although the plasma may remove most, if not all, organic species, a Cl-containing residual film often remains on the sidewall surface. The addition of a small amount of CF_4 to the O_2 or H_2/O_2 plasma can help fluorinate the embedded metallic contamination, rendering the metal complexes more soluble in the subsequent wet cleaning chemistries and DIW rinse. Residual Cl species are a concern because of possible subsequent metal corrosion if they are not removed. Chapter 7 covers plasma stripping and cleaning processes.

Wet cleaning chemical manufacturers are under constant challenges to develop more effective chemical formulations. In the early days of the semiconductor industry it was possible to use phenolic and/or hydrocarbon-based strippers to remove the photoresist [6,7], but in many cases the PER required high bath temperatures ($>120^{\circ}\text{C}$) to remove the residue. As the technologies evolved, newer solvent systems based on *N*-methyl pyrrolidone (NMP) or other polar solvents were developed. The PER compositions changed with the use of antireflective coatings and the addition of Cu to the Al metal. An important feature of the wet cleaning chemistries was that they contained low concentrations of H_2O to avoid possible metal etching during the cleaning step. Processing temperatures were usually $95\text{--}115^{\circ}\text{C}$ and process times ranged from 15 to 60 min in batch immersion or spin-spray operation.

The IC manufacturing industry has made a shift to semiaqueous chemistries [8], in particular hydroxylamine (NH_2OH)-based chemistry for Al interconnect cleaning, which was developed by wet cleaning manufacturers. Water (10–30 wt%, weight percent) can be incorporated into the solution by carefully controlling the active ingredients, such as hydroxylamine, and by adding corrosion inhibitors. The hydroxylamine (HA) cleaning processes have become a significant part of the Al-based technology with features larger than $0.13\text{ }\mu\text{m}$ and can be used in both batch immersion and single-wafer spin-spray processes. Typical HA stripping processes operate in the range of $60\text{--}75^{\circ}\text{C}$ for 10–30 min. The H_2O concentration is a critical factor for effective cleaning. If too much H_2O is added to the formulation, then metal corrosion occurs. If the H_2O concentration falls below an optimum level, the formulation will not be sufficiently aggressive to remove the PER. Water rinsing following hydroxylamine-based cleaning also causes Al corrosion [9]. This is due to the water hydrolyzing the amine during the rinsing step, forming hydroxide ions (OH^-). The hydroxide ion then reacts with Al to cause Al corrosion. An intermediate rinse step with isopropyl alcohol (IPA) or NMP is introduced before the final water rinse to overcome the water-induced Al corrosion issue. Chapter 10 thoroughly outlines the chemistry behind Al corrosion. Commercially available formulations have been developed as an alternative intermediate rinse to replace IPA and NMP, because of their lower flash point than IPA and their lower toxicity than NMP.

Fluoride-containing aqueous, both semiaqueous and solvent-based, strippers have been developed as alternatives to hydroxylamine-based cleaning products for Al interconnect applications [10]. In particular, the dilute sulfuric acid hydrogen peroxide (DSP) with HF added (DSP+) has widespread use in semiconductor device manufacturing. The disadvantage of fluoride-containing cleaning is that the fluoride etches oxide dielectric materials [2]; however, this is desirable.

4.3.1.1 Amine-Based Chemistry

For BEOL Al interconnect cleaning applications, amine-based solvent strippers commonly are used to remove photoresist polymers through penetration, swelling, and dissolution mechanisms [11]. One of the first amine chemistries used in the IC industry for photoresist stripping in the presence of metals was NMP [12]. Alkanolamine-based and hydroxylamine-based PER formulations were introduced in the 1990s [10]. The amine chemistries were and still are used mainly for PER removal. The amine chemistry PER process typically uses immersion baths because of the long process time (i.e., 15 min) and high temperature (i.e., 65°C).

The hydroxylamine-based residue removal mechanism is through reduction and complexing. Because the Al PERs are primarily metallic salts after the plasma treatment following RIE etching, they are insoluble in water, acids, and bases. These insoluble metal salts are reduced to a lower oxidation state and subsequently chelated with the ligand to form soluble metal complexes in the hydroxylamine solution [13].

4.3.1.2 Fluoride-Based Chemistry

In addition to the amine-based formulations, semiaqueous and solvent-based strippers containing fluoride also have been developed for PER removal. A common acidic composition is $\text{H}_2\text{O}/\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$ (diluted sulfuric acid and hydrogen peroxide or DSP), to which a small amount of HF is sometimes added (known as DSP+). The initial intent for this cleaning solution was for post-Al line etching using dilute H_2SO_4 and H_2O_2 or O_3 [14]. In via cleaning, the addition of a small amount of HF to the solution (known as DSP+) slightly etches the sidewall dielectric, undercutting the polymeric residue and lifting it off the surface [15]. The DSP solutions are effective at removing Al_2O_3 ; first, the H_2O_2 oxidizes the exposed Al, then, H_2SO_4 , a low-pH solution, etches the Al_2O_3 [16]. Although DSP+ can be used for post-via etch cleaning, it has limited use for post-Al line etching because of the potential for undercutting the dielectric below the line.

An important driving force for the shift to aqueous chemistries, especially those containing aggressive components such as fluoride, has been the change from batch immersion and batch spin-spray tools to single-wafer cleaning methods. Single-wafer cleaning has gained acceptance for BEOL photoresist stripping and PER removal, especially for 300-mm wafers at the <130-nm node. Batch immersion processing with F-based chemistries can be challenging because they are more likely to attack the metal and cause corrosion during post-metal cleaning. In addition, batch processing takes more time compared with single-wafer processing for the residue and photoresist to be removed from the wafer surface. Therefore, longer processing times are needed for the diffusion and removal of the F^- ions during the H_2O rinsing, to avoid a possible increase of metal corrosion. However, F-based chemistry

formulation has been successful for single-wafer processing because of the low process temperatures and short process times.

4.3.2 Copper Interconnect Cleaning

The introduction of Cu and low-k materials has changed the basic method for fabricating interconnects. Microprocessors using Cu interconnects were first introduced by IBM in the mid 1990s because Cu has a lower resistivity and far better resistance to electromigration failures than Al [17]. Chapter 10 outlines the difference between Al and Cu and their use in ICs.

The Damascene process is used for Cu interconnect fabrication, in which a silica-based dielectric insulating material is etched to create trenches and vias. For technology nodes smaller than 90 nm, a dielectric hardmask is used as the etching mask to pattern low-k materials. Next a metal barrier layer and a Cu seed layer are deposited into these features. The trenches and vias are then overfilled with Cu using electrochemical deposition and finally the excess Cu is removed by a CMP process, followed by an aqueous post-CMP cleaning step. Chapter 5 outlines the Cu dual Damascene process.

PER forms during etching of trenches and vias into the silica-based films. Upon exposure to the Cu layer at the bottom of a dual Damascene via, which contains Cu and CuO_x , plasma ion bombardment sputters these materials onto the sidewall, which mix into the PER, and are implanted into and sputtered onto the surface. The Damascene postetch cleaning process requires the removal of the remaining photoresist, the sidewall and surface residues, the sputtered metals, and the hardmask, while at the same time it must be compatible with the Cu and low-k dielectric materials. Since many low-k materials contain varying amounts of carbon, for example, the carbon-doped oxide films, an O_2 plasma stripping process cannot be used without damaging the features [18]. Therefore, liquid chemicals typically are used for trench and via PER cleaning.

The cleaning chemistries used for Al technology are inadequate for the low-k and Cu films. Many of these cleaning chemistries contain solvents with caustic components that will attack these materials. Since Cu is even more susceptible to corrosion than Al, earlier types of solvent chemistries (including hydroxylamine-based chemistries) have proved to be too aggressive. The wet chemical suppliers have responded by designing compositions with higher H_2O content (60–90 wt%). These chemistries can be acidic, mildly basic (pH 7.5–8.5) with an F-based additive, or strongly basic (pH > 11).

Most of the commercially available compositions are custom formulations with very specific concentration(s) of the active ingredient(s). Each component performs a particular function. For example, aprotic solvents are classified as molecules that do not contain an O–H bond, such as diethyl ether and acetone. On the other hand, protic solvent molecules contain a H atom attached to an electronegative atom, such as O, with the general formula ROH.

4.3.2.1 Fluoride-Based Chemistry

In Cu interconnect BEOL cleaning applications, aqueous, semiaqueous, and solvent-based strippers containing fluoride were developed for photoresist and dielectric masked PER removal. In recent years, as technology nodes advanced to 65 nm and beyond, the device dimensions became smaller and the etching mask became thinner. At these nodes, TiN (titanium nitride) metal hardmask, as opposed to photoresist and dielectric masks, is used to increase the etching selectivity between the mask and low-k materials. Formulations of various fluoride-based cleaning chemistries are shown in Table 4.3-5.

4.3.2.2 Oxidizing-Based Chemistry

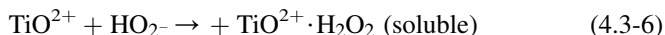
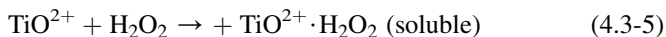
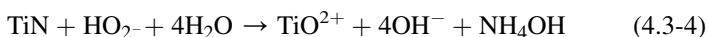
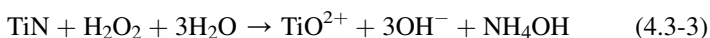
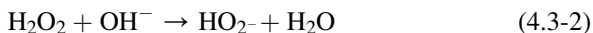
As technology nodes have advanced beyond 45 nm, achieving critical profile control of trenches and vias has become more challenging. A TiN metal hardmask is used to increase RIE selectivity to the underlying low-k materials to achieve better profile control. In addition, undercutting the hardmask can generate voids during subsequent barrier metal, Cu seed layer, and Cu filling depositions, which will cause reliability issues. The cleaning solution is required to remove the polymers from the sidewalls and via bottoms, which are generated during etching, and to be compatible with Cu and low-k materials. In addition, the cleaning solution also must be able to etch the TiN hardmask to form pulled-back/rounded corner morphology and/or completely remove the TiN [19].

Hydrogen peroxide-based chemistry, with the addition of one or more metal corrosion inhibitors, is formulated for metal hardmask PER [19–21] cleaning applications. Postetch polymer residues are removed through oxidation. The role of the metal corrosion inhibitor is to protect Cu in the presence of H₂O₂, which is added to the formulation at the point of use. Cu is passivated in the alkaline region and therefore is more easily protected by Cu corrosion

TABLE 4.3-5 Commercially Available Fluoride-Containing BEOL Postetch Cleaning Formulations
Chemicals in Formulation
NH ₄ F, H ₂ O, aprotic solvents, amines
HF, NH ₄ F, amines, solvent
NH ₄ F, amines, glycols
Quaternary fluoride, aprotic solvent, amines
HF, NH ₄ F, H ₂ O, amines, solvents
BEOL, back-end-of-line.

inhibitors in this regime. Thus these cleaning solutions are formulated most commonly with $\text{pH} > 7$. The formulations can etch TiN and remove post-RIE residue in a single process step. Chapter 10 discusses the pH regions of stability and corrosion for Cu.

The oxidizing cleaning solutions are designed to etch the TiN metal hardmask. The chemical reaction proceeds through the following steps [22]:



Hydrogen peroxide and hydrogen peroxide anion (HO_2^-) oxidize TiN to titanyl (TiO^{2+}). The titanyl further reacts with H_2O_2 and the (HO_2^-) to form soluble peroxy-titanate ($\text{TiO}^{2+} \cdot \text{H}_2\text{O}_2$) complexes. Eqs. 4.3-4 and 4.3-5 are the TiN etching rate-determining steps. The peroxide anion is the key species responsible for TiN oxidation, thus the TiN etching rate increases with increasing pH because the HO_2^- species concentration increases with increasing pH [22].

Formulations with high TiN etching rates are desirable. Thus by processing at high pH, TiN removal can use a lower temperature ($\sim 50^\circ\text{C}$), since H_2O_2 decomposes more rapidly at higher temperatures. In addition, TiN etch-enhancing compounds are incorporated into formulations, these compounds are theorized to speed up the TiN etch rate-determining steps, thus increasing the overall TiN etching rates. These TiN etching enhancers have no significant impact on the Cu etching rate, which is almost two orders of magnitude less than the TiN etching rate [21]. Fig. 4.3-1 shows scanning electron microscope pictures of the metal hardmask PERs, sidewall residue, and via bottom residues; (A) is before treatment and (B) is after treatment using oxidizing chemistry [20].

4.4 COMMON CHEMISTRIES AND THEIR APPLICATIONS

4.4.1 Organic Contamination Removal and Photoresist Stripping

The organic compound most commonly requiring removal from the wafer surface is photoresist. After being used for feature patterning, any remaining resist becomes a contaminant. Most commonly a sequence of dry stripping followed by wet cleaning is used for removing photoresist, although all-wet methods can also be used. Dry methods for stripping and cleaning are discussed in Chapters 6 and 7.

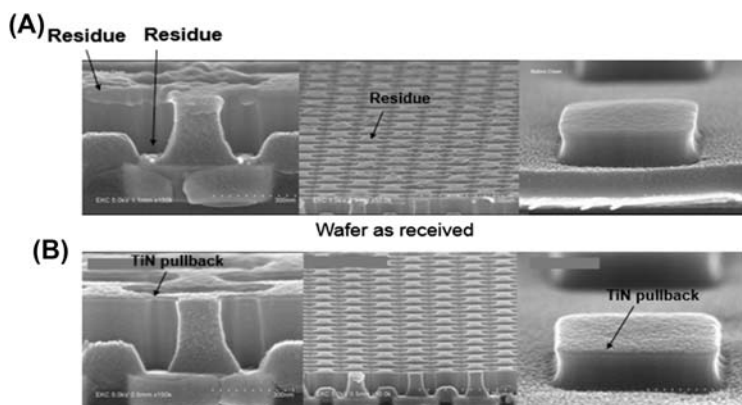


FIGURE 4.3-1 Scanning electron microscope pictures of TiN hardmask (A) before treatment and (B) after treatment with an oxidizing chemistry. The residues are completely removed and TiN pullback is minimal, but can be seen [20]. *Used with permission of DuPont/EKC.*

Photoresist compounds are composed mainly of an organic resin, the composition of which depends on the lithographic wavelength to be applied, and a light-sensitive photoactive compound, such as diazonaphthoquinone, which is photochemically transformed by light exposure to a “photoacid.” For many years the resin was novolac based; however, for 248 nm and shorter wavelengths, where absorption by novolac materials is excessively high, various chemically amplified resists have been used [23]. Currently, extreme ultraviolet lithography is nearing viability for use in manufacturing, for which new resists are being developed. For example, metallic-based resists are being proposed for high resolution, sensitivity, and etch resistance [24]. These will also require stripping solutions.

4.4.1.1 Sulfuric Acid/Hydrogen Peroxide Photoresist Stripping

The most common chemistry used for FEOL post-plasma strip cleaning, and in some cases complete stripping of resist, is SPM. The ratio may vary, but SPM is commonly a mixture by volume of between 2:1 and 4:1 H_2SO_4 (96 wt%): H_2O_2 (30 wt%) at temperatures in excess of 100°C . In some cases, this temperature is reached by the heat of mixing because of the exothermic reaction of the chemical components shown in Fig. 4.4-2. Batch spin-spray systems can provide preheating of chemicals and point-of-use mixing to achieve these high temperatures, the hardware is discussed in depth in Section 4.7.2. In other cases, a heated bath is used to maintain the temperature above 100°C .

In SPM solutions the organic material is oxidized to form H_2O -soluble carboxylic acids [5] and SPM is very effective for removing photoresist and other organic residues. However, extensive rinsing is required to remove the

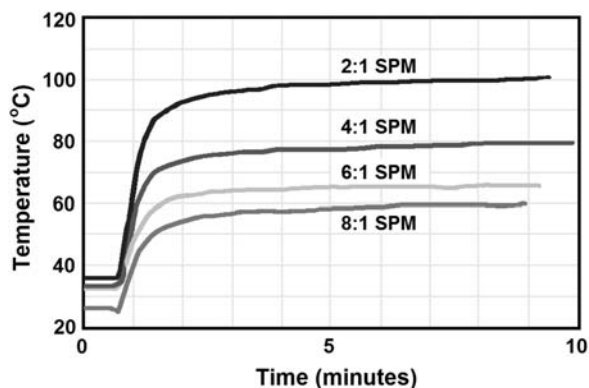


FIGURE 4.4-2 The temperature change of various concentration SPM solutions upon mixing the H_2O_2 . The temperature of H_2SO_4 initially was at 20°C . The reaction is exothermic and the equilibrium temperature is reached in less than 5 min. *Courtesy of TEL FSI International.*

viscous chemical and avoid hygroscopic sulfate residues on the wafer, which, over time, can absorb moisture or chemical contaminants from the atmosphere and nucleate small particulate defects [25,26] or haze [27]. To mitigate this effect, small amounts of HF can be added to the mixture to alter the Si or SiO_2 wafer surface wetting properties, causing the chemical to efficiently drain from the wafer after rinsing [27]. SC-1 treatment after SPM is the most commonly used countermeasure, and is quite effective.

High-dose implants, used to form device source and drains in the Si substrate, are also implanted into the photoresists that protects the unimplanted areas. This leads to the formation of a carbonized “crust” layer of the photoresist that is a challenge to remove by wet processing only. However, very high-temperature SPM processes have proven successful at removing such implanted resists and have been implemented in mass production. Typically, the H_2SO_4 is heated to a temperature well in excess of 100°C , then mixed with H_2O_2 close to the dispense point to take advantage of the exothermic reaction. In one method, steam is added to the SPM at point of use in a closed chamber to further accelerate the stripping rate and minimize chemical consumption [28]. Typical SPM blends at 2:1 ~ 4:1 ratios have high water content (H_2O_2 being nearly 70% water). Although the mixing of H_2SO_4 with water leads to a beneficial exothermic reaction, it also dilutes the chemical. Use of steam enables high temperature and high energy with minimal dilution.

The SPM chemistry, if applied to a bare Si surface, creates a thin oxide layer. Oxide layers created by wet chemistries are known as chemical oxides. Such oxides typically grow in a self-limiting fashion (oxidizing species lacking sufficient energy to diffuse through the formed oxide layer to react at the Si surface) such that they do not exceed about 1.0–1.2 nm in thickness,

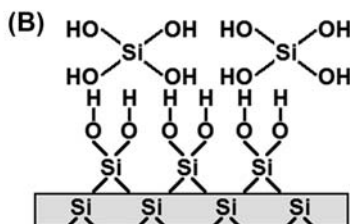
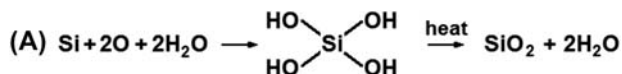


FIGURE 4.4-3 A suggested model for the chemical SiO_2 formation in an oxidative aqueous medium. (A) Formation of silicic acid molecule and SiO_2 after a thermal treatment and (B) chemical oxide with a highly hydrated SiO_2 , or aggregates of silicic acid clusters. *Adapted and reproduced by permission of ECS—The Electrochemical Society from J. Guan, G. Gale, G. Bersuker, Y. Jeon, B. Nguyen, J. Barnett, M. Jackson, D. Burkman, P. Peavey, K. Yokomizo, K. Ueno, M. Yamasaka, M. Imai, S. Kitahara, N. Shindo, Sixth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing, T. Hattori, R.E. Novak, J. Ruzyllo (Eds.), The Electrochemical Society, 99-36, Pennington, NJ, 1999, pp. 77.*

and do not comprise the stoichiometric SiO_2 formed by thermal oxidation [29,30]. As shown in Fig. 4.4-3, it has been theorized that chemical oxides comprise aggregates of molecular silicic acid clusters whose hydroxyl groups ($-\text{OH}$) can be removed by heating, causing the film to densify [31,32], further discussed in Chapter 9. Although chemical oxides have been reported to have dangling bond defect densities on the order of $1.0 \times 10^{12}/\text{cm}^2$ at the $\text{Si}-\text{SiO}_2$ interface [33], about two orders of magnitude higher than thermally grown oxide, their quality is understood to be dependent on the type of solution in which they are generated [34], for example, SPM generates a different chemical oxide than SC-1.

4.4.1.2 Ozonated Deionized Water Photoresist Stripping

The impact of the SPM cleaning and required rinsing chemicals on the environment, safety, and health (ESH) represents a serious drawback. Furthermore, as is often the case, ESH and cost are related, and the heavy use of chemicals and H_2O results in high cost to the industry. Consequently, many efforts have been made to replace SPM with more environmentally benign (and lower cost) processes using only O_3 and DIW. The strong oxidizing power of O_3 , and even stronger hydroxyl radical ($\cdot\text{OH}$) formed by decomposition of O_3 in H_2O , make such a process attractive. However, the solubility of O_3 gas in H_2O is about 20 ppm at room temperature ($\sim 20^\circ\text{C}$) [35], which is considered low, and its rapid consumption (the reaction rate is limited by diffusion of fresh O_3 to the reaction surface) presents a fundamental limitation. Taking advantage of the inverse proportionality of the solubility with

temperature, a subambient temperature process was developed to increase the concentration of O_3 dissolved in an immersion bath [36]. Although the process is effective in some cases, it has not been used widely since the same low-temperature condition that enables increased O_3 solubility simultaneously limits its heterogeneous reaction rate with photoresist at the wafer surface. To avoid the limitations of O_3 immersion batch processing, several researchers have developed processes that instead use spray or vapor H_2O in the presence of O_3 gas [37–39]. Using this technique, the O_3 must only diffuse through a thin liquid layer or can absorb into the H_2O mist at the wafer surface, making it possible to simultaneously have a high temperature and a high O_3 concentration. However, the high-dose implanted PR strip has proved very difficult to strip effectively with DIW/ O_3 alone, and inorganic components such as metals in the residues require H_2SO_4 for complete removal. Tomita et al. pointed out that the solubility of O_3 in H_2SO_4 is both higher and less temperature dependent than in H_2O , and found SOM solutions more effective than DIW/ O_3 for removing hardened resist without ashing [40].

4.4.1.3 Organic Contamination Removal

The SC-1 solution, introduced in Section 4.2, is capable of removing light organic contamination by oxidation, in addition to particle removal, to be discussed in the next section. The SC-1 solution is not used for photoresist stripping, but can be used as a clean-up step to remove low-level organic residues. Typically, after SPM, SC-1 is used for its particle removal capability.

4.4.2 Particle Removal

4.4.2.1 SC-1 Particle Removal

State-of-the-art wafer cleaning processes accomplish particle removal by a combination of chemical and physical methods. The SC-1 step of Kern's RCA sequence, used for batch immersion processing, historically has been the workhorse chemistry for particle removal, and although alternatives have been proposed, it remains so. The SC-1 solution consists of NH_4OH (29 wt%): H_2O_2 (30 wt%):DIW at a pH near 10 [1–3]. The original formulation was 1:1:5 by volume at 70–80°C, although modifications have been made, which will be discussed in Section 4.5.1. SC-1 removes particles by a combination of slightly etching the SiO_2 or Si to undercut particles and electrostatic repulsion of the particle and the surface because of the high pH of the solution. The electrostatic repulsion is a result of electrostatic double layer phenomena, which will be briefly introduced here. More thorough treatments of this topic can be found in Chapter 3 and in Ref. [41].

The etching of Si in SC-1 is accompanied by (1) the formation of a thin chemical oxide (as with SPM) and then (2) the removal of the oxide layer. Hence, the combination of Si oxidation to form SiO_2 and then the subsequent

etching of SiO_2 results in Si removal. For leading-edge processes with tight material budgets, such Si loss is increasingly undesirable and this has led to a reduction in concentration and temperature to mitigate the loss.

The interaction of a solid surface with a surrounding electrolyte, including adsorption of ions from the solution and/or dissociation of surface groups, causes the surface to become charged. The layer of counter-ions (called the Stern layer) that attach to this charged surface and move with it and a diffuse layer of ions outside it are together termed the electrical double layer. The charge at the boundary between these two layers is called the zeta potential, which determines how surfaces will interact in the liquid. Surfaces of opposite charge will be attracted to one another and surfaces of like charge will repel one another. Electrical double layer and zeta potential concepts were first applied to the elucidation of wafer cleaning mechanisms by Riley et al. [42].

Zeta potential is a function of both the pH and the ionic strength of solution. The zeta potential decreases from a positive value at very low pH to a negative value at high pH, as the OH^- ion concentration increases. The pH value for which the zeta potential is zero is known as the isoelectric point. Figure 3.3-4 in Chapter 3 shows the zeta potential of particles of various materials, as a function of pH, obtained by electrophoresis measurements. For high pH values in the SC-1 regime of 9–10, these materials take on strongly negative charges. Therefore, a silica particle or silicon nitride particle, for example, is repelled from a silicon oxide wafer surface in SC-1. This electrostatic interaction prevents deposition of particles from solution onto wafers, and is important in ensuring particles that have been detached from a wafer during cleaning do not simply reattach.

One alternative to SC-1 is the use of choline (trimethyl 2-hydroxyethyl ammonium hydroxide) mixed with H_2O_2 [43]. Like NH_4OH , choline is alkaline and is capable of etching Si, and has similar zeta potential characteristics.

4.4.2.2 RCA Cleaning and HF Particle Removal

Hydrofluoric acid, applied after an oxidizing chemistry such as SPM or DIW/O_3 , has been studied as an alternative to SC-1 for particle removal. The undercut and lift-off mechanism can be achieved by DHF. However, the zeta potential conditions are less favorable than with SC-1 because of the low pH. It has been shown that detached particles in the chemical layer close to the wafer surface can redeposit during rinsing. The use of anionic surfactants to adjust the zeta potential and thus help to avoid attraction can be an effective countermeasure for improving the net particle removal in DHF [44].

Menon et al. [45] conducted a study of particles added by various steps of an RCA clean. Fig. 4.4-4 shows the number of particles, in the size range of 0.2–0.5 μm , added during each step of an RCA cleaning sequence (SC-1–SC-2–10:1 BHF), and for the entire clean [45]. The SC-2 (acid rinse) and BOE

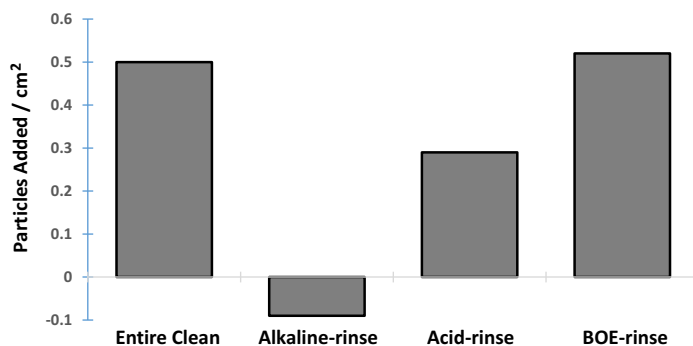


FIGURE 4.4-4 Particle addition to a wafer after various steps in a wet cleaning process. The alkaline-rinse process is SC-1 and the acid rinse is SC-2 [45]. Note that the entire cleaning process particle addition is approximately equal to the particle adds from the buffered oxide etch (BOE) process. *Used with permission from Solid State Technology.*

treatments added particles to the wafer, whereas the SC-1 (alkaline rinse) treatment removed a few particles. The overall number of particles added to the wafer was very close to that added during the HF-last (BOE) process. The addition of the particle contributions from the individual steps of the cleaning sequence does not equal the particles added for the entire clean. This implies that some of the particles added during the initial cleaning step are removed during the next step, possibly because of liquid–wafer interfacial effects or chemical dissolution [46,47]. Furthermore, both SC-1 and the SC-1–SC-2 sequence leave the wafer hydrophilic, and the surface tensions of both chemical solutions are similar [48]; hence, the differences in particle removal behavior must be attributed to chemical interactions at the liquid–wafer interface.

Use of HF solutions as a final cleaning step generally has been known to add particles to wafers [47,49]. Stripping a SiO₂ film from a Si wafer renders the surface hydrophobic. When the Si wafer is immersed in the DIW rinse, the hydrophobicity causes particles in the DIW to be preferentially deposited on the wafer surface. Hence, particle levels on wet etched wafers in immersion systems generally correlate well with particle levels in the rinse tank, but not the HF bath [45]. In Section 4.7.4 various rinse tank configurations to minimize particle deposition are discussed.

The hydrophobic or hydrophilic nature of a Si surface can be quantified by the contact angle of a drop of H₂O on that surface. A surface with a contact angle of 0 degree is completely hydrophilic, whereas a contact angle of equal to or greater than 90 degrees (H₂O beading on the Si surface can have an angle greater than 90 degrees) defines a completely hydrophobic surface. The contact angle of H₂O with an HF-etched surface is approximately 66–85 degrees, whereas that of buffered HF on a hydrophobic surface is around 70 degrees

TABLE 4.4-6 Effects of Cleaning Chemistry on Si Surface Contact Angle [52]		
Bare Si Wafer After Cleaning With Chemistry	Contact Angle Mean and Standard Deviation	Hydrophobic or Hydrophilic
DI H ₂ O	16 ± 1.3	Hydrophilic
SC-1 1:1:5 + Rinse	10 ± 0.8	Hydrophilic
SC-2 1:1:5 + Rinse	9 ± 0.5	Hydrophilic
BOE 1:10 + Rinse	66 ± 2.1	Hydrophobic
Measurements of contact angles are typically performed with a goniometer. Wafers are assumed to be hydrophilic before the cleaning sequences are performed. Reprinted from Microcontamination Magazine, Tom Cheyney, Los Angeles, CA, Copyright 1990. Used with permission.		

[50,51]. The Si surface wettability of buffered HF is not significantly different from that of H₂O. To improve the wettability of the etchant, Kikuyama et al. [50] recommend the use of hydrocarbon surfactants. Their studies have shown that the use of carefully selected surfactants can significantly reduce the particle contamination in wet etch processing. The application of surfactants in chemical solutions, especially for pre-gate oxide cleaning, should be approached very carefully. Surfactant molecules are potential sources of carbonaceous residues. Aggregates of these molecules, called micelles, are in the submicron-size range and can represent an added source of particles. Also, many surfactants cause foaming, which can leave stains on the wafer surface and cause uneven etching.

The hydrophilic/hydrophobic nature of wafer surfaces after various chemical-cleaning steps is noted in Table 4.4-6 [52]. The SC-1 and SC-2 treatments leave the wafer very hydrophilic. Kern [53] reported that SC-1, at ~80°C, etches SiO₂ at a rate of about 0.5 Å/min, whereas SC-2 shows very little change in thickness. The etching in the alkaline SC-1 solution undercuts SiO₂ beneath particles, allowing them to be more easily dislodged from the wafer. Also, Niida et al. [54] have shown that the zeta potential of common types of particles in alkaline solutions is significantly more negative than in acidic solutions. However, in SC-2 solutions, the zeta potential of most particles is positive. The zeta potential of the Si surface is typically negative in most solutions (isoelectric points of Si and SiO₂ surface are 1.5–4.0). Hence, SiO₂ particles tend to be attracted to the Si wafer surface in SC-2, whereas they are repelled in SC-1 solutions [54].

4.4.2.3 Particle Removal and Surface Roughness

Mishima et al. [55] investigated the effect of different NH₄OH concentrations in SC-1 in a batch immersion tank on particle removal. They recommended

that the SC-1 solution should have a lower NH_4OH content (0.50–0.05 times lower than conventionally used) to improve the particle removal capability by a factor of two, without any increase in surface roughness. Ohmi et al. [56] and Meuris et al. [57] also studied the effect of NH_4OH concentration on Si surface roughness and particle removal capability. Ohmi recommended a 0.05:1:5 SC-1 as the optimum mixture based on particle removal and etching rate studies. The optimum NH_4OH concentration is correlated to the solution etching rate in which the particle removal efficiency (PRE) is the highest. The roughness resulting from 0.05:1:5 at 0.2 nm Ra (average roughness) is half that at 1:1:5, as measured by atomic force microscopy. Increased surface roughness is found to cause a decrease of the gate dielectric breakdown charge (Q_{BD}), as shown in Fig. 4.4-5 [56]. Meuris et al. [57] also showed that an increase in Si surface roughness from an SC-1 solution could directly cause a decrease in capacitor yield. They recommend a 0.25:1:5 mixture of SC-1 for good particle removal without significant surface roughness. The difference in optimum NH_4OH concentrations between Ohmi's and Meuris's studies has been reported by Meuris to be due to a difference in solution temperatures. Ohmi's studies were conducted at approximately 80°C, whereas Meuris's work was at 70°C.

In addition to surface roughening, minimizing silicon and silicon oxide material loss has become increasingly important as critical dimensions shrink. Kirkpatrick et al. [58] explain several important effects in sub-32-nm complementary metal oxide semiconductor FEOL. Especially critical are the implanted lightly doped drain and dielectric material sidewall spacers around the gate. Harmful device effects from Si loss include altering effective channel length, modification of implant depths or spacing, and interference with strain engineering because of altered distance between the gate edge and the raised

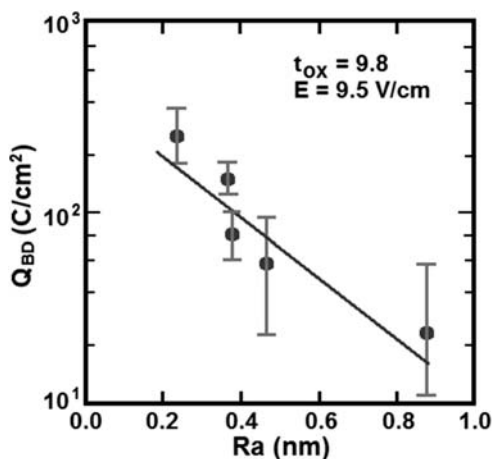


FIGURE 4.4-5 Effects of Si surface roughness (Ra) on gate dielectric breakdown charge (Q_{BD}) [56]. The thickness of the gate oxide is 9.8 nm and the voltage applied to the devices is 9.5 MV/cm. Courtesy of TEL FSI International.

SiGe (silicon germanium) channels. These effects have a direct impact on key device performance parameters such as source/drain resistance and on and off current of the device ($I_{\text{on}}/I_{\text{off}}$).

Various concentrations and dilutions have been proposed for SC-1. In some cases, concentrations used for single-wafer processors are different from those of batch immersion tanks, especially for particle removal. Also different engineers have found optimum concentrations for their specific processes. Other researchers have proposed cleaning sequences that avoid SC-1 entirely. One published method uses 1 wt% HF and 1 wt% HCl in a continuous flow batch immersion system [59]; other concentrations and formulations are also used. Alternative sequences will be discussed in [Section 4.5.3](#).

4.4.2.4 Megasonic Particle Removal

As early as 1979, researchers at RCA reported using megasonic energy to remove particles from wafers [60]. The use of megasonic energy is very effective at removing particles, in combination with H_2O alone or with SC-1. As Menon and Donovan [61] have pointed out, the force required to remove a particle from the wafer surface must equal or exceed the force of adhesion. They state that the force required to remove a particle is a function of particle size, particle and wafer surface composition, and the nature of the liquid medium [62]. For a silica particle of 1- μm diameter with a mass of 5×10^{-13} g that is adhered to a bare Si surface, the van der Waals force of adhesion in H_2O is approximately 4×10^4 dyn [63].

Megasonic cleaning is a subset of ultrasonic cleaning, where ultrasonic refers to sound waves at frequencies above the range of human hearing (greater than about 20 kHz). The megasonic range of frequency is the most commonly used for particle removal from semiconductor wafers, because lower frequency ultrasonic cleaning is known to lead to damage even to relatively large structures. Commercial megasonic wafer cleaning systems typically operate at a frequency of 700 kHz–1 MHz.

Megasonic energy is transmitted through transducers, which are piezoelectric crystals. In batch immersion systems these are usually mounted at the bottom of the tank. When configured in this manner, the sonic waves travel through the liquid in a direction parallel to the wafer surface. In some single-wafer systems, megasonic rods have been used in which the piezoelectric crystal is mounted at the end of the rod and the energy transfer to the wafer is perpendicular to the wafer surface [64]. However, today's single-wafer systems mostly do not use megasonics.

A sonic wave is a pressure wave, inducing alternating periods of compression and rarefaction in a liquid. When the intrinsic “tensile strength” of the liquid is exceeded during the low-pressure phase, the liquid pulls apart to form a cavity. Dissolved gas in the liquid diffuses into the cavity and, through a process known as rectified diffusion, the cavity increases in size over

repeated cycles. This phenomenon, known as cavitation, leads to a host of interesting and useful effects. It is generally accepted that the motion of cavities leads to particle removal via localized fluid flows very close to the wafer surface (the boundary layer in megasonic cleaning being very thin relative to typical immersion systems) and formation of shock waves upon cavity collapse [65]. These flows near the wafer surface are a type of acoustic streaming, which is an important phenomenon in megasonic cleaning.

Acoustic streaming refers to time-independent motion of liquid due to viscous attenuation. Three types of streaming phenomena are known to exist in solutions subjected to sound fields, namely, Eckart streaming, Rayleigh streaming, and Schlichting streaming. Eckart streaming, which is mainly bulk flow, is characterized by vortices much larger than the wavelength of sound, whereas Schlichting and Rayleigh streaming result in liquid flow close to a solid boundary with length scales much smaller and on the order of sound wavelength, respectively. In acoustic cavitation, bubbles formed during the low-pressure cycle of the sound wave either continuously oscillate over many cycles (stable bubbles) or grow in size and eventually collapse in less than a few cycles (transient bubbles). The flow generated from oscillating bubbles, known as microstreaming, is considered to be instrumental in particle removal, whereas shock waves and microjets generated from transient bubbles are believed to be responsible for damage [66].

Megasonic cleaning is a very effective method for removing deep submicron-sized particles from Si wafers, particularly when used in conjunction with SC-1 solutions so that the physical mechanisms of megasonics are combined with the chemical mechanisms (surface undercutting and zeta potential repulsion) of SC-1. When SC-1 chemistry is used in lieu of H_2O , along with megasonic energy, the PRE is substantially increased. In one study [67], SC-1 solutions are found to consistently produce high cleaning efficiencies and relatively small variability between runs, whereas the lowest cleaning efficiencies and largest variabilities are seen with DIW. [Section 4.4.2.6](#) discusses methods for performing PRE testing.

Megasonic cleaning has been used in the industry for many years; however, for recent device generations problems with pattern damage have arisen. From approximately the 130-nm technology node onward, patterns (in particular polySi gate structures) have become small enough that they can be damaged by megasonic cleaning. Controlling the uniformity of the energy distribution in the system is one of the keys to mitigate such damage. [Fig. 4.4-6](#) shows an example of a damaged polySi line caused during the megasonic cleaning process, possibly because of cavitation and subsequent collapse of the bubble.

Much work has been performed to minimize the damage to sensitive features on the wafer surface, with various modifications being proposed to the processes and the equipment (both batch and single wafer) for optimizing particle removal versus damage. Christenson [68] and Vereecke [69,70] have reported on optimization of SC-1 megasonic processes for high PRE and low

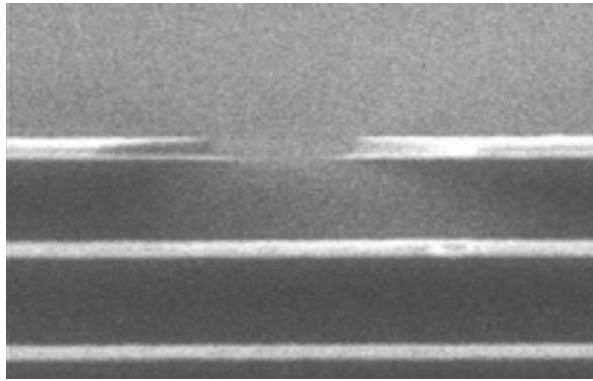


FIGURE 4.4-6 A damaged polySi line possibly because of cavitation caused by megasonic high energy [71]. *Courtesy of TEL FSI International.*

feature damage. Christenson has improved the configuration of the megasonic transducers in a batch immersion system to produce uniform PRE across the wafer, alleviating previous areas that had low PRE, and studied the addition of gases to the cleaning solution [71]. Vereecke [69,70] has also characterized the addition of various gases, such as O_2 and N_2 , into the SC-1 solution. Earlier work [72] has shown that dissolved gas, such as air, O_2 , H_2 , He, and Ar, in cleaning solutions has an effect on cleaning efficiency. The addition of the O_2 and N_2 , dissolved in SC-1 cleaning and H_2O rinsing solutions, indicates that the PRE increases. Kang et al. [73] studied the effect of various dissolved gases on particle removal and damage in DIW with megasonics. They found the addition of H_2 gas led to not only the largest increase in PRE but also the greatest increase in damage because of the effect of H_2 on increasing transient cavitation. Potential effects of the liquid type on damage were studied by Franklin, who reported that lower damage is obtained from megasonics when using organic solvents [74].

Measuring the emission of light from multiple bubbles as they collapse in the liquid is a method developed by Ferrell [75] for monitoring megasonic energy. The phenomenon of “multibubble sonoluminescence” is used to map the collapsed bubbles. This is especially useful for determining the uniformity of the process, because images of removed particles can be compared with the location of the collapsed bubbles to increase the efficiency of megasonic cleaning.

Other proposed enhancements to megasonics include pulsing and the use of variable frequencies [76,77].

Although extensive research has helped clarify fundamental mechanisms of megasonics in recent years, continual shrinking of patterns has made the physical damage problem increasingly acute. Although improvements have been made to processes and equipment, both batch and single wafer,

applicability of megasonics in current leading edge manufacturing is focused on blanket (unpatterned) films or patterns of relatively large dimension.

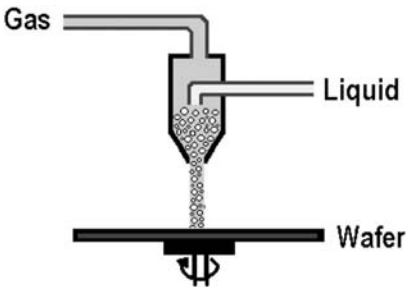
4.4.2.5 Fluid Jet Particle Removal

Nozzle-based cleaning was introduced specifically for single-wafer cleaning, both horizontal and vertical wafer orientation. Nozzle-based cleaning has been commercially available on CMP tools for some time [78]. The use of specialized nozzles, orientations, and configurations for removal of small particles has proliferated since the introduction of 45-nm node processes, especially for low-cost devices such as DRAM (dynamic random access memory, used for general data storage and requiring power to hold the data) and NAND (the most common type of non-volatile memory, which can store data without power and is widely used for portable devices) memory IC devices. An overview of some of the nozzle technologies for cleaning is presented in Table 4.4-7. As can be seen from this table, many researchers are investigating the nozzle particle removal and extending the technology to other single-wafer processing such as resist stripping, as discussed in Section 4.4.1. Variations in the application of the H₂O stream to the wafer can be accomplished by any of these nozzle types and wafer orientations. What differentiate these methods are their cleaning efficiency and damage imparted to the IC devices. The typical nozzle used for pressurized spray is shown in Fig. 4.4-7. The ejector nozzle uses a pressurized gas, typically nitrogen and sometimes clean dry air, which is mixed with DIW, thus the name mixed fluid jet, which form droplets of the liquid that are sprayed onto the wafer surface. The supply pressure of the liquid mainly controls the size of the ejected H₂O droplet, whereas the pressure and flow rate of the gas controls the droplet velocity [79] and to some extent the droplet size [80]. Fig. 4.4-8 shows the size range of

TABLE 4.4-7 Overview of Aqueous-Based Nozzle Technology for Particle Removal

Cleaning Technique	Description	Reference
Sonic-pulsed spray	A piezoelectric crystal is used to apply sonic energy to the H ₂ O stream at the point of use	[84]
Mixed fluid jet spray	Mixed fluid gas, typically N ₂ , is added to H ₂ O near the point of use. The N ₂ gas nucleates the H ₂ O into droplets that impact the surface causing particle removal	[79,81]
Pulsed fluid jet spray	H ₂ O is passed through a pump, which delivers a pulsed spray onto the wafer. Pulsing allows particle removal at lower energy	[82]

FIGURE 4.4-7 Schematic of mixed fluid (liquid + gas) nozzle-based cleaning based on pressurized gas spray [79]. *Used with permission of Karen Reinhardt.*



particles from a nozzle and their respective velocity [81]; typically, the particle size is less than 60 μm . Chemicals other than H_2O have been used for cleaning efficiency improvement and to reduce the amount of damage to sensitive features [83]. The use of HF is proposed to undercut particles and remove them by lift-off, in addition, to the physical effect imparted by the pressurized spray [83]. Fig. 4.4-9 shows the PRE using the mixed fluid jet spray with various chemicals, including H_2O , SC-1, and HF.

4.4.2.6 Particle Removal Monitoring

PRE is a measurement used for comparing cleaning processes with respect to a wafer with a large load of particles on the surface. The equation to calculate PRE is:

$$\text{PRE} = [(P_{\text{pre}} - P_{\text{post}})/P_{\text{pre}}] \times 100\% \tag{4.4-7}$$

where P_{pre} is the number of particles on the wafer before processing and P_{post} is the number of particles on the wafer after processing. There is no industry

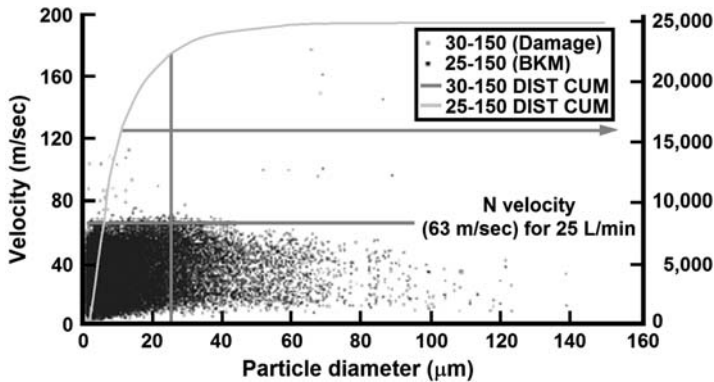


FIGURE 4.4-8 Distribution of particle sizes with mixed fluid jet cleaning [81]. Note that the damage occurred at velocities greater than 30 m/s and particle sizes greater than 150 μm . *Dist Cum* is cumulative distribution and *BKM* is best known method. *Used with permission from Solid State Technology.*

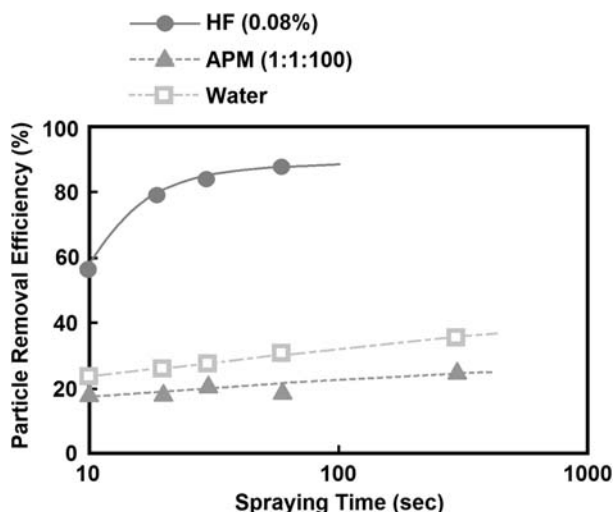


FIGURE 4.4-9 Graph showing particle removal efficiency for HF-last Si surfaces using mixed fluid jet spray with a variety of cleaning chemicals [83]. The N_2 flow rate is 80 L/min, APM is ammonium hydroxide and hydrogen peroxide mixture (SC-1 at 1:1:100 $NH_4OH:H_2O_2:H_2O$). Reproduced by permission of ECS—The Electrochemical Society from H. Hirano, K. Sato, T. Osaka, H. Kuniyasu, T. Hattori, *Electrochem. Solid State Lett.*

standard for creating these particle monitors. Many different methods and particles, typically Si_3N_4 , SiO_2 , or PSL are used at sizes ranging from >100 to <40 nm.

Typically, particles are spun onto the wafer via a liquid solution in a controlled manner or sprayed on the wafer using a dry deposition system that provides well-controlled and localized deposition [85]. Next, the wafers are analyzed for the number of particles on the wafer, P_{pre} . The particle level on the wafer should not exceed the maximum number of defects the inspection tool is able to recognize. Typically, the number of particles are about 8000–10,000 on a 300-mm wafer because higher precounts may make the wafers easier to clean (some particles may be agglomerated, loosely adhered, etc.) and this results in artificially high PRE. Chapter 12 discusses the defect analysis tools used for the inspection process. The wafer is then processed using the cleaning and surface preparation method to be tested. Finally, the wafer is analyzed for the number of particles on the wafer after the process, P_{post} , and PRE calculated. Fig. 4.4-10 shows an example of PRE, also known as cleaning efficiency [86].

4.4.3 Metal Contamination Removal

A useful approach to categorizing types of metal contamination, a critical step in designing strategies for their removal, has been provided by Morinaga and

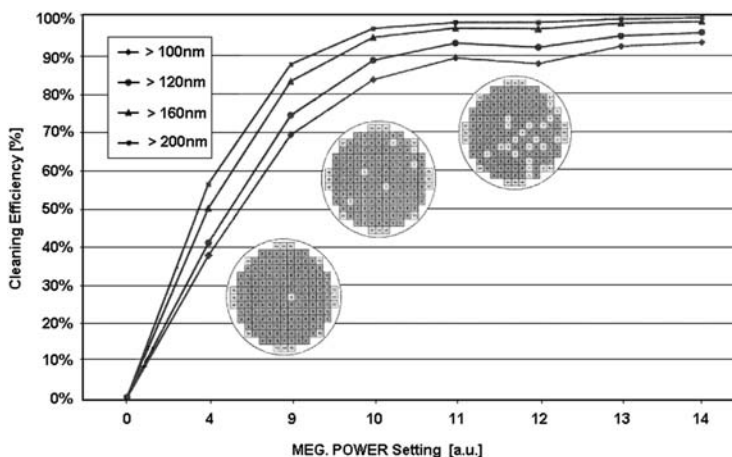


FIGURE 4.4-10 Particle removal efficiency and a comparison to the damage to various sized features from cleaning at different megasonic power levels. The wafer maps show increasing damage (lighter colored die) and increasing particle removal efficiency with increasing megasonic power [86]. Used with permission of Alexander Lippert.

Ohmi [87]. Chapter 2 also discusses types of metallic contamination classifications. Type (I) is defined as electrochemical deposition, type (II) as hydroxide precipitation, and type (III) as film inclusion. An example of the first type would be metals of a higher electronegativity than Si (Cu is an often cited example) depositing via redox (reduction–oxidation) reactions onto bare Si in HF [87].

The second type of metal deposition occurs in SC-1 solutions. It first should be pointed out that SC-1 is capable of removing some contaminant metals. Aqueous ammonia will form complexes with Co^+ , Co^{2+} , Cu^{2+} , Fe^{2+} , Ni, and Zn [5]. It has been reported that Ni and Zn contamination after APM are inversely proportional to the NH_4OH concentration [88]. Other metals, however, are known to be deposited from solution onto Si wafers. Metals in SC-1 can form metal hydroxides, which easily attach to silanol groups on chemically oxidized wafer surfaces; Fe, Al, Ni, and Zn form especially stable metal hydroxides [89,90]. Consequently, just 1 ppb of Al in SC-1 solution can result in about 1.0×10^{12} atoms/cm² deposited on an immersed wafer [91]. This is type (II) metal contamination in the categorization of Morinaga and Ohmi. Type (III) can also occur in SC-1 when metals are embedded in the chemically grown oxide. Since these metals need to be controlled to $<1.0 \times 10^{10}$ atoms/cm², an additional cleaning step is needed to remove them.

The SC-2 solution, consisting of HCl (37 wt%): H_2O_2 (30 wt%):DIW, is commonly employed just after SC-1. It should be noted that a rinse step

always takes place after the chemical cleaning step. A range of temperatures from near ambient to 70°C or more may be used. Hydrochloric acid is effective because it increases the oxidation strength of the solution in addition to complexing transition metals such as ions of Fe [5].

Hurd et al. [92] found that diluted HCl can be sufficient to remove metal contaminants normally encountered in FEOL cleaning processes. They further explained that keeping the dilution below 0.12 M is beneficial in avoiding particle deposition from the acidic solution.

Hydrofluoric acid is another logical reagent for metal removal since, in addition to its low pH, it etches oxide and can thus be effective for metals embedded in chemical oxides. IMEC researchers developed an alternative process to the RCA clean that uses dilute HF after an oxidizing step to remove metal contamination very effectively [93]. However, care must be taken when using HF because some metals in the solution, most notoriously Cu, can readily deposit on the Si surface. IMEC extensively studied the behavior of metals in HF cleaning, including the effects of processing under illumination versus darkness [94,95]. They concluded that, although electrochemical effects cause Cu to plate out onto Si when exposed to light, the addition of HCl to DHF is an effective countermeasure, provided a sufficient HCl concentration is used. At low concentrations, added chloride ions actually increase Cu deposition by catalyzing the deposition reaction, whereas at high concentrations, complexation of Cu ions by Cl^- prevents deposition [93].

Adding H_2O_2 to dilute HF has been proposed for Cu removal simultaneously with oxide removal, although it also etches the Si surface slightly because of the combined presence of an etchant and oxidizer [87,96,97]. Finally, DIW/ O_3 solutions have also been shown to be effective at removing certain metals such as Cu and Fe [98].

4.4.4 Cleaning Process Sequences

Although modifications to temperatures and concentrations have been proposed, the classical SPM—DHF—SC-1—SC-2 cleaning sequence, described in Section 4.2.1, remains the most commonly used sequence for cleaning steps before oxidation, annealing, and deposition. Alternatives have been used to some degree, the two most well known were independently developed by researchers at IMEC and at Tohoku University.

The IMEC cleaning sequence [93] was proposed after extensive research to elucidate the mechanisms of the RCA clean, in an effort to provide a lower cost and less environmentally deleterious process, with a simpler process sequence. The initial SPM step is replaced with $\text{H}_2\text{SO}_4/\text{O}_3$ or DW/O_3 for organic contamination removal and chemical oxide growth. Next, a DHF or DHF/HCl step is performed to remove metals and also particles by undercutting the chemical oxide, resulting in a hydrophobic surface. If a chemical oxide-passivated surface is preferred, an optional DIW/ O_3 step can be

included as a last step before drying, into which some HCl can be added to prevent metal deposition. IMEC researchers have reported excellent metal decontamination results, as well as reduced Si consumption, using this sequence [93].

A group under the leadership of Professor Tadahiro Ohmi at Tohoku University proposed another alternative cleaning sequence that is performed entirely at room temperature [96]. A DIW/O₃ solution (5 ppm) is first used to remove organic contamination as well as some metal contamination and to create a chemical oxide. Next, an HF/H₂O₂/H₂O solution, with an ionic surfactant and megasonic energy added, is used for the removal of particles, metals, and the chemical oxide formed by the initial DIW/O₃ step. A DIW/O₃ rinse, also with megasonic energy, follows this chemical step. The final sequence is a 0.1 wt% DHF step followed by a DIW/megasonic rinse [96,99].

SONY researchers proposed a sequence suitable for single-wafer processors, in which very short oxidizing and etching steps, of only a few seconds, are alternated until the surface has been cleaned. The oxidizing is accomplished using DIW/O₃ and the etching is accomplished by 1 wt% DHF. Thus particles and metals can be removed in a short process time, without the need for megasonic energy, which can damage fine patterns [100]. Most single-wafer processing, however, still utilizes variations of the classical sequence. The most prominent change has been to shorten the time for each step, to avoid unreasonably long process times with corresponding low throughputs for single wafer. Batch immersion processes have historically had SC-1, SC-2, and rinse times on the order of 5–10 min, whereas in single-wafer spin tools these steps have been reduced to times ranging from 1 min down to 30 s or less. Also, the sequences have been simplified such that a full SPM–DHF–SC-1–SC-2 sequence is relatively infrequent in state-of-the-art single-wafer processes. Sequences now more typically include two or a maximum of three chemical steps. Typical examples are SC-1–SC-2 for precleaning, and SPM–SC-1 for postcleaning, depending on the specific application. In memory technologies using single-wafer, DHF and SC-1 combinations are common.

4.5 PROCESS VARIABLES AFFECTING CLEANING

Wet cleaning and etching processes rely on chemical reactions that remove material and/or contaminants from the wafer surface. Such reactions are functions of time, temperature, and concentration. To determine the best process conditions there are many considerations to be taken into account and balanced. These include tolerable amounts of thin film and silicon removed from the wafer surface, surface morphology, cycle time and throughput, environmental effects, and cost. In general, once optimum process conditions have been established, it is desirable that they be maintained with maximum stability and control, so that every wafer is subjected to identical conditions.

4.5.1 Concentration and Temperature Effects

The original SC-1 solution is 1:1:5 ($\text{NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$) at 70–80°C, conditions deemed suitable to the cleaning and surface conditioning task at that time. With earlier critical dimensions much larger than today's, there was no concern about the amount of oxide etched by the solution and no concern about angstrom-level effects on surface morphology. And, indeed, for approximately 20 years there was no cause for such concern. Eventually it became clear, however, that concentrated SC-1 solutions at high temperature causes microroughening of the Si surface (particularly, when used immediately after HF), and it was found that this could affect device performance [101–103]; Chapter 9 discusses Si surface morphology in detail. To alleviate the problems, more dilute, lower temperature SC-1 solutions were proposed. Various researchers showed that, especially with the introduction of megasonic energy to enhance particle removal, such solutions could be as effective as the higher temperature, more concentrated solutions while mitigating the microroughening effect [101,104,105]. It was proposed that about 2.0 nm of substrate etching, in the presence of megasonic energy, was sufficient to undercut and remove embedded particles, whereas the zeta potentials were not substantially altered by the relatively small change in pH. Thus the original 1:1:5 $\text{NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ ratio migrated to considerably more dilute ratios [104,105]. Currently, SC-1 ratios about 10 times more dilute than the original RCA formulation are common. In fact, more dilute solutions are shown to be beneficial because the “thickness” of the electric double layer, which defines the distance of interaction between charged solid surfaces in solution, is inversely proportional to the square root of the solution's ionic strength [41]. Therefore, in more dilute solutions the repulsion at high pH acts over a longer distance in the solution.

The introduction of more dilute SC-1 solutions means the volume of H_2O_2 and NH_4OH used by semiconductor factories are significantly reduced. Therefore, environmental and cost benefits also result. Subsequent in-depth study of the role of H_2O_2 in SC-2 solutions revealed that concentrated solutions could be replaced by simple dilute solutions of HCl in H_2O [106,107], or at least by SC-2 solutions of significantly lower concentration and temperature [105].

In high-temperature SC-2 solutions, the H_2O_2 decomposes so quickly that the concentration of H_2O_2 can decrease to near zero in as little as 30 min [106]. Conditions of about $1.2 \times 10^3 \text{ M}$ HCl at 50°C are recommended by IMEC and Texas Instruments researchers [92]. Owing to the volatility of the chemical components, SC-1 and SC-2 solutions are easily removed from wafer surfaces without leaving residues. However, as discussed later, control of solution concentration becomes a challenge for the process engineer.

In terms of process control, the most critical aspect of SC-1 is that it will etch SiO_2 , and that it can consume Si at rates that depend on the relative

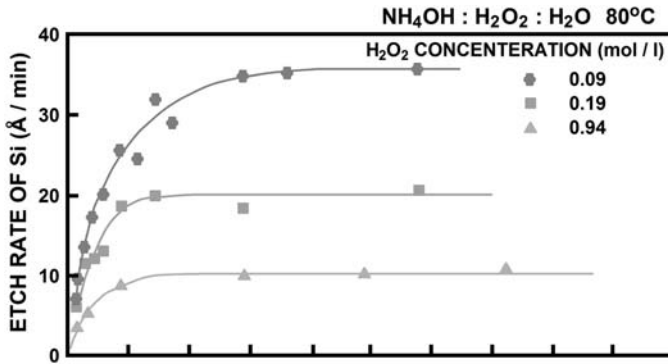
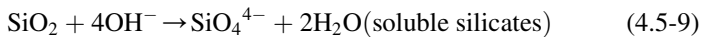
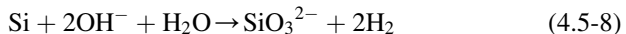


FIGURE 4.5-11 Silicon etching rate as a function of the concentrations of NH_4OH and H_2O_2 in an SC-1 solution [109]. Reprinted with permission from H. Kobayashi, J. Ryuta, T. Shingyouji, Y. Shimanuki, *Japanese Journal of Applied Physics*, vol. 32, 1993, pp. L45.

concentrations of NH_4OH and H_2O_2 . An alkaline solution will etch Si along the $\langle 111 \rangle$ crystal planes in the absence of a sufficient amount of oxidizer, resulting in roughening of $\langle 100 \rangle$ surfaces [108]. The removal rate of Si is directly related to the concentration of OH^- in the solution [109]. The reactions involved are [105]:



As the ratio of NH_4OH to H_2O_2 increases, the OH^- concentration will increase. Care must be taken to avoid conditions in which excessive Si etching and roughening may occur [109]. It was found that the etching rate increases linearly with NH_4OH concentration up to a point of saturation, after which it no longer increases, as shown in Fig. 4.5-11. The maximum etching rate is inversely proportional to the H_2O_2 concentration, suggesting that the H_2O_2 creates a passivating oxide [109].

4.5.2 Time and Aging Effects

For IC manufacturing, baths of concentrated chemicals are recirculated through particle filters for various time periods, ranging from about 2 h up to several days, depending on the chemistry and application. Changes to the solutions take place over this time, and these are of constant concern to the process engineer. Such changes include:

- “Dragout” of chemical solution, caused by the adhesion of the chemicals to the wafer surface with each batch of wafers processed, causing the bath volume to reduce over time, thus requiring replenishing of the chemicals.

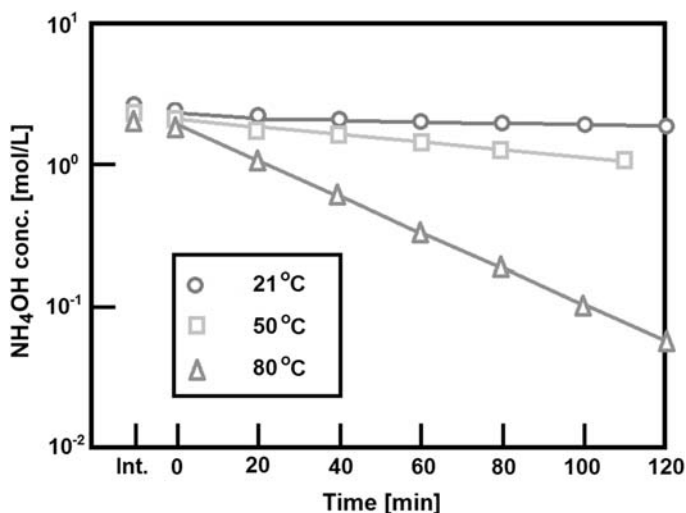
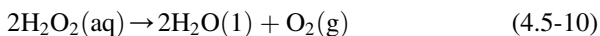


FIGURE 4.5-12 The decrease of NH_4OH concentration under different temperature conditions in an SC-1 bath with an initial 1:1:5 ratio (Int.) [110]. Reprinted with permission from I. Takahashi, H. Kobayashi, J. Ryuta, M. Kishimoto, T. Shingyouji, *Japanese Journal of Applied Physics*, vol. 32, 1993, pp. L1183.

- Different evaporation rates for different chemical constituents causing relative concentrations to change over time. An example would be H_2O evaporating faster than HF , causing a concentration increase over time. Fig. 4.5-12 shows the decrease in NH_4OH concentration in an SC-1 bath [110].
- Breakdown/decomposition of one or more chemical component, either naturally or by reaction with contaminants or etched films. A classic example is SC-1, in which H_2O_2 decomposes according to the following equation [111].



Metal contaminants in solution, especially Fe, have been found to catalyze the decomposition [112].

- Buildups of particulate, metallic, or organic contamination in the baths after many wafers have been processed.
- Unwanted precipitation, such as from surfactants or buffering agents.
- Absorption of atmospheric gases, such as N_2 , CO_2 , and O_2 , especially in recirculated baths. Carbon dioxide can affect the pH of baths because of formation of carbonate ions.

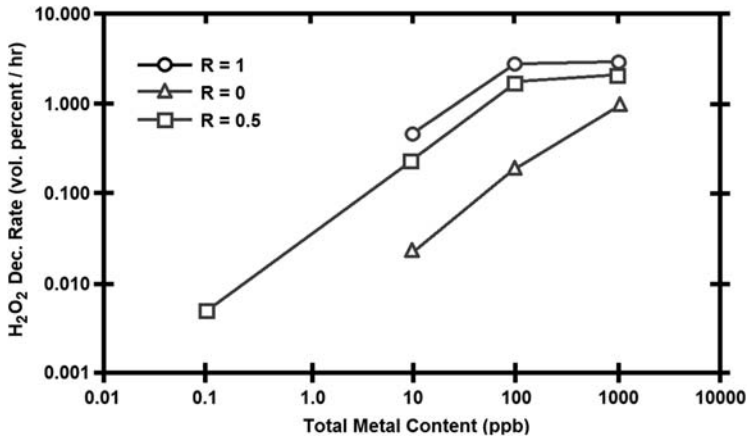


FIGURE 4.5-13 The decomposition rate of H_2O_2 in a 1:1:5 SC-1 bath as a function of Fe and Cu concentration (as well as their relative concentrations), indicating the higher dependence on Fe concentration, where $R = 0$ [113]. R is the ratio of the concentration of Fe to the total concentration of Fe and Cu. Used with permission of Ara Philipossian.

The case of SC-1 is of interest because of the relationship between bath conditions and Si surface etching and roughening, leading to subsequent effects on device performance. Researchers have presented an explanation of the reaction mechanisms, including metal (Fe) ion aggregates creating local H_2O_2 depletion and galvanic corrosion due to noble metals deposited in the prior HF step [108]. In any case, metal contaminants catalytically accelerate the decomposition of H_2O_2 [113], as shown in Fig. 4.5-13.

Chemical solutions that use H_2O_2 decompose rapidly; hence, they need to be replenished with fresh H_2O_2 before a new batch of wafers is processed. Typical bath life for SC-1 or SC-2 mixtures operating at temperatures from 50 to 80°C is few hours before replenishment is needed [110]. Thus chemical usage costs can be very high.

As a result of concentration variability over time, state-of-the art immersion systems often employ in-line chemical concentration monitoring systems, and “chemical spiking” pumps to periodically refresh consumed chemical components of the bath. Chapter 11 discusses the methods for concentration and component analysis. A further trend, consistent with the use of increasingly dilute aqueous chemistries and single-wafer spin tools, is to use single-pass without recirculation, so that the process will not be affected by chemical concentration and purity changes over time. Concentrated solutions such as BHF, H_3PO_4 , and others, however, remain recirculated, mainly because of the high cost of those particular chemicals. A unique feature of hot phosphoric acid (H_3PO_4) is that aging can actually be beneficial because the buildup of etch products in solution improves the selectivity of Si_3N_4 versus SiO_2 etching. This occurs because of the oxidized silicon in solution causing the etching rate

of SiO_2 to be strongly suppressed (an etching rate of essentially zero can be achieved) with only a relatively mild decrease in Si_3N_4 etching rate. This has led to proposals for intentionally spiking the bath with Si-based additives to improve selectivity. Excessive concentrations of such additives, however, will lead to increased particulate contamination, especially at lower temperatures [114]. Another key concern with phosphoric acid baths is that depletion of the H_2O in the 85 wt% H_3PO_4 solution causes a decrease in the Si_3N_4 removal rate.

4.5.3 Effects of Sequence

The most common cleaning sequence for pre-thermal processing is the RCA sequence. The sequence variation with the most profound effect is the “HF-last” process, where HF is the last step before final rinsing and drying. Such HF-last processing results in a hydrophobic or nonwetted, H-terminated Si surface [115] as opposed to the hydrophilic or readily wetted chemical oxide, hydroxyl-terminated surface associated with the standard cleaning sequence, with SC-2 as the last process, for example. Common HF-last applications include:

- Surface preparation before epitaxial Si, SiGe, or Ge growth.
- Cleaning and surface preparation before polySi deposition where Si–Si contact is needed.
- Cleaning and surface preparation before metal (e.g., Co, Ni) sputtering to form metal silicides.
- Cleaning etched contacts before contact liner deposition.

Such applications are very sensitive to small amounts of chemical or native oxide. Therefore, not only is HF-last cleaning required, but also often the queue time between the cleaning step and the subsequent process must be short to avoid native oxide regrowth.

An alternative RCA sequence is reversing the SC-1 and DHF steps; the sequence of the last three steps is SC-1–DHF–SC-2, resulting in an interesting effect. Although the process is not HF-last, chemical oxide thickness as measured by an ellipsometer is closer to the value measured after HF-last than to the amount grown by SC-1. Since HF-last should have no chemical oxide, the “film” that is being formed is discussed in Chapter 9. The overall electrical oxide thickness of integrated gate oxide stacks is also similar for HF-last and HF + SC-2 last processes, whereas for SC-1 + SC-2-last it is on average 0.05–0.1 nm thicker [32].

4.5.4 Deionized Water Quality

Because H_2O is used in every aqueous cleaning process, the impact of H_2O quality on defect levels is considerable. Contaminants are removed from the

H₂O before use in the cleaning process. Bacteria growth in ultrapure DIW must be avoided; ultraviolet (UV) irradiation, ozonation, and filtration are methods used to kill and remove the bacteria. The sequence of UV light use, ozonation, and filtration can also affect the final particle count [116,117]. The UV light breaks bacteria into smaller fragments, which manifest themselves as particles. One bacterium can generate several fragments.

Shadman et al. [116] show that the total organic carbon (TOC) level dropped dramatically when the membrane filter is placed before the UV unit. The filter removes particles larger than a certain size, whereas the remaining organic particles are oxidized and broken down by the UV unit. In the reverse sequence, where the first step is H₂O exposed to the UV unit, the smaller fragments generated are more difficult to remove in the filter, thus resulting in a higher TOC level, as shown in Fig. 4.5-14. In spite of these research observations, most semiconductor factories still favor the use of a final filter before pumping H₂O into the fab.

Even with filtering, particles are found in the DIW supply. New 300-mm IC manufacturing facilities are striving to achieve concentrations of <0.2 particles/mL for particles $\leq 0.05 \mu\text{m}$ in DIW [118], as compared with ASTM and SEMI F-63 standards of <0.5 particles/mL for particles $\leq 0.05 \mu\text{m}$ in DIW. Studies have been performed to determine how particles in the DIW supply adhere to the wafers. Riley and Carbonell [42] show that the number of added particles does not necessarily increase with immersion time; however, the number of particles added to a hydrophilic wafer depends on the zeta

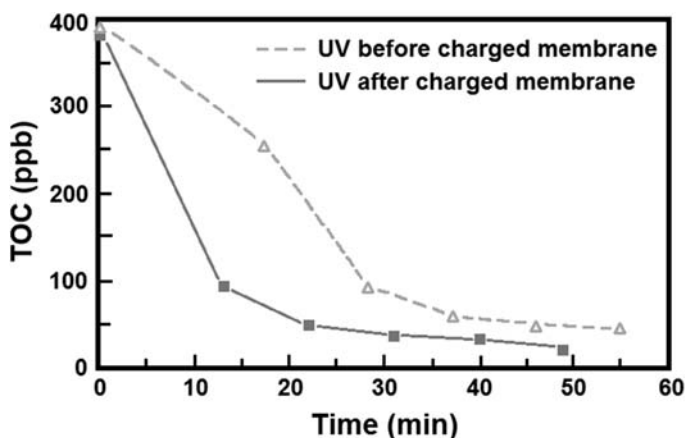


FIGURE 4.5-14 The effect on total organic carbon (TOC) of sequencing ultraviolet (UV) with a charged membrane filter; 15.2 L/min [116]. Copyright 1990 by IEST. Used by permission. Reprinted with permission from 1990 IES Proceedings, published by IEST. www.iest.org (847) 981-0100. From: F. Shadman, R. Governal, A. Bonner, *Interactions between UV and membrane filters during removal of bacteria and TOC from DI water*, in: *Proceedings of Institute of Environmental Sciences, Annual Technical Meeting, IEST, Arlington Heights, IL, 1990*, pp. 221.

potentials of the particles and wafers and their ease of wetting in H_2O . In all cases, the number of particles added to a hydrophilic Si wafer surface increased with particle concentration in H_2O .

To achieve the goal of a low particle density in DIW, multiple-pass filtration is performed using coarse and fine filters, with continuous or periodic ozonation and UV light to break down bacteria. The final filter in the finishing loop is usually a charged membrane filter. Manufacturing facilities use large quantities of DIW, principally in wafer-cleaning tools, varying from <40 to >200 L/min.

Many particles in DIW are colloidal species that are usually negatively charged, especially colloidal silica and lipopolysaccharides [119]. The use of a positively charged modified membrane filter aids in removing these negatively charged particles. Because the nature of particle capture is related to the attraction of colloidal particles of charge opposite to that of the membrane medium (often termed “electrokinetic adsorption”), these filters are very efficient at removing particles much smaller than the pore size of the membrane. Many commercial charge-modified filters are available.

Research at the University of Arizona has shown some interesting properties of charge-modified filters. Jan et al. [119] have extensively characterized the electrokinetic properties of these filters and found discrepancies between the claims of suppliers and the actual performance. The researchers measured the zeta potential, surface charge density, and saturation capacity of two different charge modified Nylon membrane filters. They found that suppliers are claiming particle filtration at pH values of H_2O ranging from 3 to 10, whereas their results showed that these filters become negatively charged (and hence ineffective) at pH values below 7.6. The wider the pH range in which the zeta potential is positive, the more efficient is the filter; nonetheless, these charged filters have shown, in field use, to be very efficient in small-particle removal from H_2O .

4.6 RINSING AND DRYING

Although a great deal of engineering goes into removal of particulate, metallic, and organic contaminants without deleterious effects on wafer materials and structures, the overall cleaning process is only as effective as the last steps, rinsing and drying. If these steps are not well engineered, all gains made in the chemical cleaning process can be lost. Rinsing between chemical steps is relatively straightforward, quench and remove the bulk chemical. Most critical, however, are the final rinsing and drying steps.

4.6.1 Evaluation of Rinsing Performance

The purpose of rinsing is to remove chemicals and replace them with DIW so that no residual chemicals remain on the wafer. Rinsing between chemical steps prevents unwanted reactions such as the formation of ammonium

chloride (NH_4Cl) that can occur in the absence of sufficient rinsing between the SC-1 and the SC-2 steps. Typically, in the final rinsing step before drying, an in-line resistivity measurement is used to ensure that chemical has been sufficiently replaced by H_2O . Any residual chemicals, because of their ionic nature, will reduce the resistivity.

The large amounts of H_2O used, as much as 2.0×10^7 gal or more per month for a fully operating wafer fab [120], makes rinsing an environmental and cost issue. It is desirable to optimize the rinsing process so that the amount of H_2O used is minimized while the effectiveness of its use is maximized. Since resistivity measurements typically are taken where the bulk rinse H_2O is going to drain, they do not necessarily represent what is occurring at the wafer surface. Mass spectroscopy techniques are useful for detecting light elements on wafer surfaces after rinsing, such as F that might remain after HF processing [121].

The final rinsing process poses different challenges depending on whether the wafers are hydrophobic (after HF) or hydrophilic (after oxidizing chemistries). In the former case, if it is desired that the Si surface remains H-terminated, often O_2 is removed from the rinse H_2O to prevent reoxidation of the Si. Also, in this case the highly active Si-H surface can easily attract contaminant particles, so rinse cleanliness and flow pattern are important. It should be noted that hot H_2O should never be used for rinsing Si after HF, since it will roughen the bare Si surface by accelerating the Si etching rate because of the higher temperature [122].

The use of $\text{O}_3/\text{H}_2\text{O}$ also allows the formation of a thin native oxide layer, called a chemical oxide, which passivates the wafer surface, thus preventing subsequent contaminant deposition [121]. If a passivated chemical oxide, SiO_x , surface is preferred after HF cleaning, it is necessary to include a final step, such as injecting O_3 into the rinse water to readily form this clean chemical oxide. Several suppliers now offer an ozonized rinse as a part of their wafer cleaning equipment. In the case of the hydrophilic surface, which is most common since sequences typically end with SC-1 and SC-2, care must be taken to avoid adsorption of metal contaminants from the final rinse water. At near neutral pH, certain periodic table column II alkaline earth metals, such as Ca and Mg, which are particularly detrimental to gate oxide integrity (GOI), can be easily deposited on oxidized Si [123,124]. It has been proposed that a competitive adsorption mechanism leads to some metal cations depositing when the concentration of H^+ cations is insufficient. Therefore, lowering the pH by addition of HCl or HNO_3 to the H_2O , for example, mitigates the metal adsorption [123,124].

At the end of a rinsing operation, a hydrophilic wafer exiting the rinse tank carries with it a thin film of H_2O about 20 μm in thickness. The boundary layer thickness during wafer pullout, on the other hand, is around 2500 μm [125]. This implies that the carryover film is entirely within the boundary layer. Transport of residual ions or particles is slower within the boundary layer than in the bulk rinsing medium. When the boundary layer is much larger than the

carryover thickness, there will be ionic and particulate species that are not rinsed off the wafer surface even after a long rinsing period.

Routine correlation of particle counts in the DIW within overall H₂O quality is useful in determining the cause of particle fluctuations. Chapter 12 discusses methods for monitoring particles in H₂O and in chemicals. For HF-etched wafers, variations in the TOC level in the rinse tank can cause major changes in the number of particles added.

4.6.2 Drying Methods and Processors

Major categories of drying methods are as follows:

1. Spin drying, both batch and single wafer
2. IPA vapor-based drying, both batch and single wafer
3. Liquid IPA drying, single wafer

4.6.2.1 Spin Drying

The earliest means of drying wafers was by simply spinning them in a centrifuge, at rates of 800–1000 rpm or more. This still is practiced for both batch and single-wafer configurations, although other methods have been introduced that provide better performance especially for more critical applications (such as HF-last). Disadvantages of spin drying include particle generation caused by the mechanical motion and flow dynamics. In addition, depending on the machine design, a centrifugal rinse may result in the generation of micrometer- and submicrometer-size H₂O droplets, which may not be effectively removed in the dry cycle. In batch spin systems a dry-only mode adds less particles and watermarks than a combined rinse and dry. As a general practice, it is recommended that the rinsing process be accomplished in the rinse tank, not in the spin dryer. An efficient dryer in the dry-only mode can add less than one particle ($<0.2\ \mu\text{m}$) onto a 150-mm-diameter wafer. Spin drying in an environment purged by dry hot N₂ can mitigate other dryer-generated defects, such as watermarks after HF processing.

For batch spin-spray processors or single-wafer spin tools, the drying is done in situ after the chemical processing step and rinsing. For wet benches, a separate module is used. The key challenge is to perform the process, which typically involves moving parts, without particle generation. This is particularly difficult with batches of 300-mm wafers. In addition, in such a batch of spinning wafers the breaking of one wafer (for example, caused by a preexisting stress crack) can contaminate the entire batch.

4.6.2.2 Isopropyl Alcohol–Based Drying

Another method of drying uses a cloud of IPA, which, because of its low surface tension, displaces water when it condenses on the wafer surface. In

addition, the high vapor pressure of IPA enables it to readily evaporate, leaving dry wafers. Drawbacks of such dryers, besides using large amounts of IPA, include the necessity of special fire safety measures and concerns about organic residue that may remain after drying.

IPA vapor dryers were introduced to achieve watermark- and particle-free wafer drying. In IPA dryers, rinsed wafers are exposed to IPA vapors, either by direct displacement using condensed IPA (the vapor dryer) or by surface tension. These dryers have fewer moving parts and tend to have fewer particle problems than centrifugal spin dryers. However, these dryers are much slower than the spin dryers and use large amounts of a liquid with a low flash point; for IPA the flash point is approximately 13°C for 80–100 wt% solution. Ohmi et al. [126] reported that the H₂O content in the IPA, the IPA heating system, and the IPA vapor velocity are major variables affecting the performance of a vapor dryer.

An effective drying technique that uses smaller amounts of IPA than the IPA vapor dryer relies on the so-called Marangoni effect. Specifically, if a gradient of surface tension exists between two liquids, there will be a flow from the higher to the lower surface tension region. This explains why high alcohol content drinks such as brandy can be observed to seemingly crawl up the inside of a glass in opposition to gravity. In wafer drying, an IPA vapor is introduced into a chamber above a DIW bath in which the wafers are immersed. The IPA (low surface tension) diffuses into the H₂O (high surface tension) at the surface, so there is a gradient of surface tension between the surface (having a lower surface tension because of the IPA) and the bulk H₂O below (having a higher surface tension). As the wafers are pulled slowly upward through the interface, H₂O is pulled away from the surface by the Marangoni effect [127]. This method offers advantages that have led to its wide use. First, it uses less IPA volume than the IPA vapor method, which makes its cost lower and its safety requirements less stringent. Second, it can be carried out at much lower temperatures. Because H₂O is physically removed, it is easier to avoid watermarks. In addition, this approach is claimed to remove particles from the wafer surface more effectively than is possible with IPA vapor dryers [128].

Surface tension drying referred to as “Marangoni drying” was developed by researchers at Phillips Research Laboratories [129] and further explained in Refs. [130,131]. As the wafers are removed from the rinse tank at speeds of 1–2 mm/s, the air/H₂O/Si interface is exposed to the vapors of an H₂O-soluble organic liquid, typically IPA. This IPA is absorbed into the H₂O at the interface, and the IPA/H₂O interface induces a surface tension gradient (STG) within the curvature of the interface, causing the H₂O meniscus to contract. This effect allows the H₂O to sheet off a smooth hydrophilic wafer surface. The technique leaves a hydrophilic Si wafer surface drier than a spin drying operation, as measured by the amount of H₂O evaporated from the surface after the completion of the drying operation. Fig. 4.6-15 is a schematic of an

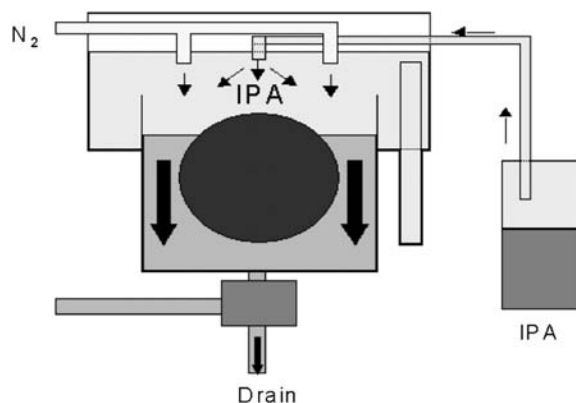


FIGURE 4.6-15 Schematic of isopropyl alcohol (IPA) surface tension gradient dryer. *Courtesy of TEL FSI International.*

immersion surface tension dryer. Other potential liquids that have low surface tension and form volatile vapors, such as diacetone alcohol and 1-methoxy-2-propanol [132], have been found to be potentially suitable for Marangoni drying, but in practice are not used.

Owing mainly to issues of stiction-related pattern collapse, drying by dispensing 100% IPA liquid onto a rotating wafer, displacing water, then spinning off the IPA has become commonplace. Typically the IPA is dispensed from a nozzle above the center of the rotating wafer, and the nozzle is subsequently scanned toward the wafer edge as the wafer rotates and water is displaced. Control of temperature and humidity is important for avoiding condensation of liquid on the dried surface [133].

4.6.3 Defects Related to Rinsing and Drying

4.6.3.1 Watermarks

Historically, hydrophobic surfaces and mixed hydrophilic and hydrophobic patterns after HF-last processing have presented the greatest challenge for drying. A type of defect called watermarks, water spots, or drying stains has become well known in the industry [134–138]. Examples of watermarks are shown in Fig. 4.6-16. The mechanism of formation of this defect involves Si oxidation, dissolution of the formed oxide, and the dynamics of the drying. The oxidation occurs because of dissolved O_2 either in the H_2O or diffusing into the H_2O during rinsing. Subsequently the oxide is dissolved into the H_2O according to the overall sequence [136]:



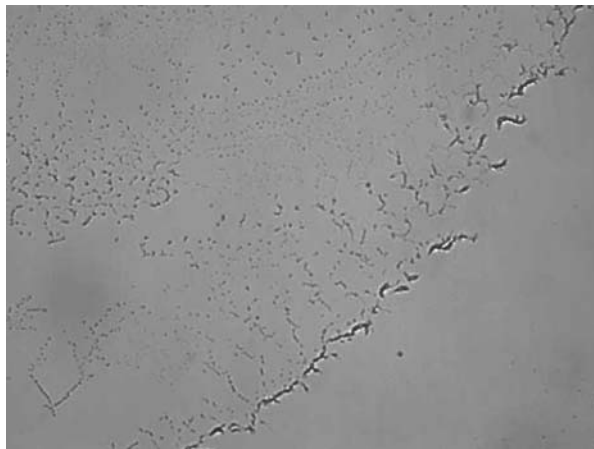


FIGURE 4.6-16 Examples of watermarks on a dried wafer. *Used with permission of Karen Reinhardt.*



As the wafer is dried, the dissolved residue can precipitate to form defects containing Si and O. One way this happens is if evaporation of the H_2O film or H_2O droplets occurs, since solid material will not evaporate. Therefore, the H_2O must be completely removed. Mixed hydrophobic and hydrophilic areas are particularly sensitive, most likely because while the H_2O is instantly pulled away from the hydrophobic areas, the hydrophilic areas can trap it. So, with intricate patterns the H_2O can be trapped in contact with bare Si, and the above-mentioned reactions occur. Because such watermark defects tend to kill an entire chip when they occur, avoiding them is a critical concern for the process engineer. IPA-based drying techniques have become favored over spin drying for sensitive applications. In general, strategies for mitigating watermarks include:

- Performing the HF step in the same chamber where rinsing and drying occur, to avoid transfers through air between HF cleaning, rinsing, and drying steps.
- Eliminating O_2 from the rinse H_2O .
- Eliminating O_2 from the ambient atmosphere in the dryer (e.g., by N_2 purging).
- Ensuring residual fumes from chemicals such as HF or ammonia have been completely purged from the chamber before wafer drying.
- Using IPA-based drying instead of spin drying.

- Avoiding deposition of H₂O droplets (e.g., splashing) on dried portions of the wafer.
- Avoiding evaporation of the H₂O at the transfer steps or when the wafer is idle.

A type of drying defect is identified that has similarities to watermarks but is formed by a somewhat different mechanism. In cases where the HF step is performed in the same chamber as rinsing and drying, a watermark-like defect with a much higher density of silica/silicate residue can form if HF vapor is not adequately removed from the chamber before rinsing and drying. In conjunction with the oxidation and etching mechanism described earlier, the HF accelerates the etching portion, leading to a thicker residue containing fluoride [139]. It also is found that heavily doped polySi is more sensitive than undoped Si for forming this type of defect [140]. More recent work has noted a mechanism of defect formation in IPA drying because of the following mechanism. During rinse, residual ammonia and HF in the chamber ambient form ammonium fluoride that dissolves into the rinse water. As the rinse water is replaced by liquid IPA, the ammonium fluoride forms precipitates because it is not soluble in the IPA [141].

4.6.3.2 Pattern Collapse

Stiction-related pattern collapse has become the singlemost troublesome problem for wafer drying, and perhaps for cleaning in general, in recent years. This has happened because of patterns becoming simultaneously taller and thinner (higher aspect ratio) and spacing between the patterns decreasing as scaling has advanced. Subsequently, the surface tension of liquid between fine patterns can be sufficiently high to pull the features toward one another, causing them to collapse and stick together, called stiction. This phenomenon has led to the investigation of new drying methods.

Fig. 4.6-17 depicts a liquid between two pattern lines that are being pulled together because of what is known as the Laplace pressure (ΔP) and the surface tension force at the contact line (F_x) [142]. These forces are represented as follows.

$$\Delta P = P_{\text{atm}} - P = (2\gamma \cos(\theta - \phi)) / (d - 2\delta) \quad (4.6-14)$$

$$F_x = \gamma \sin(\theta - \phi) \quad (4.6-15)$$

Here, $(d - 2\delta)$ represents the gap between patterns as they are being pulled together. A smaller gap leads to a larger Laplace pressure. Another point that is clear from the equations is that the forces are directly proportional to the surface tension, γ . Therefore, replacing the water between patterns with a lower surface tension liquid, such as IPA, has been a successful method for mitigating collapse. The collapse force has been compared against the pattern strength as a function of Young's modulus of the material in various studies [143].

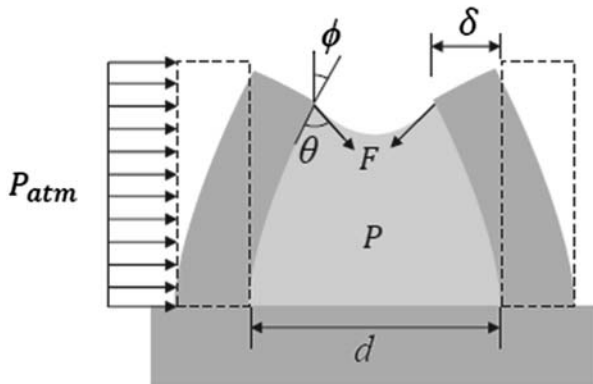


FIGURE 4.6-17 Schematic of a liquid between two pattern lines that are being pulled together because of Laplace pressure and surface tension force. *Used with permission of Glenn Gale.*

For very high aspect ratio structures with very tight spacing, even liquid IPA may not be capable of completely avoiding collapse. For such cases, applying chemicals to make the surface hydrophobic, which causes the Laplace pressure to approach zero in the equation, has been proposed by Watanabe et al. [144]. For Si-based patterns a silylating agent can be used for the surface modification. Watanabe et al. used trimethylsilyldimethylamine.

For the most extreme cases of high aspect ratio and narrow spacing, where neither liquid IPA nor hydrophobic surface modification treatments are effective to avoid collapse, an ultimate solution of supercritical CO_2 drying has been proposed [145]. Above the critical point, a fluid has solvating properties of a liquid but negligible surface tension and low viscosity like a gas. Therefore stiction forces should be absolutely minimal. Carbon dioxide (CO_2) is the most commonly proposed fluid for this purpose, with a critical point of 30.8°C and 73 bar. However, this is expected to have high cost and there are additional concerns such as managing contamination in the high-pressure process. Chapter 8 discusses supercritical processes.

4.7 AQUEOUS CLEANING EQUIPMENT

Commercially available aqueous-based cleaning, rinsing, and drying equipment used for the manufacturing of IC devices is discussed in this section. Many types of equipment exist to carry out these processes.

The choice of material for the construction for cleaning equipment is important: the material must stand up to the exposure of chemicals and H_2O , and must not degas, shed particles, corrode, or decompose during the life of the tool. Components and subassemblies must also be able to withstand constant chemical exposure. Piping for DIW must be free of additives or contaminants that can elute into the fluid and must be compatible with oxidizing

chemicals such as H_2O_2 , O_3 , and hot H_2O [122]. Examples of material used for aqueous processing equipment and components are perfluoroalkoxy (PFA), polytetrafluoroethylene (PTFE), and oxy-1,4-phenylene-oxy-1,4-phenylene-carbonyl-1,4-phenylene, also known as polyetheretherketone (PEEK). Polyvinylchloride (PVC) has been shown to shed particles and thus is rarely used for aqueous tool components. PEEK is the material that is recommended for DIW transport systems because of its smooth surface properties, lower leaching potential, and temperature insensitivity; however, it is very expensive. Stainless steel and other metal components must be avoided; stainless steel has the potential to add metallic contaminants into the H_2O stream.

4.7.1 Immersion Wet Benches

Batch wet-chemical cleaning systems are either immersion based or spin spray based. Immersion wet stations consist of one or more chemical baths, rinse tanks, and dryers. A wafer cassette is first immersed in the chemical bath for a specified length of time and then transported to the rinse tank. High-productivity immersion tank systems are fully automated with robotic transfer of the wafer cassettes from the 300-mm front opening unified pod to the carrier used to process the wafers through the system. In some cases, the robot may hold the wafers throughout the entire process with a specially designed arm (“cassette-less”) or the robot may transfer the wafers to specially designed cassettes that minimize the touch points on the wafer surface, thus minimizing shielding of the wafer surface, the volume of chemical displaced by the cassette, and the amount of chemical carried over to the rinse tank. Materials used for the carrier and the robot must be compatible with all of the chemicals used in the wet bench. Cross-contamination of the robot must be avoided, and thus, most immersion wet bench systems are designed with a robot cleaner.

Immersion baths can reuse the chemical solutions for several batches of wafers. Using the same chemical for a long time leads to a buildup of contamination [146]. Recirculation with point-of-use filtration is essential to keep the particles in the bath to an acceptable level. To control particle levels, the time allowed for recirculation, filtration between wafer loads, and starting with a fresh batch of chemicals is very important. If adequate time is not allowed for the particle level in the bath to decrease by filtration, gradual buildup of particles will rapidly cause the bath to be ineffective. [Section 4.7.6.2](#) will discuss aspects of point-of-use filtration.

Typically, the immersion process is repeated for each chemical in the sequence, in separate tanks, with rinses being performed in separate rinse tanks after the preceding chemical process. Rarely are the same rinsing tanks used for different chemicals in the wet bench.

Drying is performed in a separate vessel in the wet bench. Immersion systems have also been designed to serially process the RCA sequence,

including the intermediate rinses and STG drying in the same processing tank. Many commercial configurations are available and most companies will custom design an immersion tank system to a specific cleaning sequence used for manufacturing.

Megasonic capability is available in most immersion batch systems and is typically used in combination with SC-1 cleaning. Although a single-wafer immersion-cleaning system using megasonic transducers has been described [147], immersion cleaning has not seen wide use in single-wafer systems.

4.7.2 Centrifugal Spin-Spray Batch Systems

In a spin-spray processor, a cassette or cassettes of wafers are placed in a bowl and then rotated on a shaft at a certain speed while chemical solutions are sprayed on the wafers. After the chemical spray is stopped, the wafers are rinsed with a DIW spray, and then spun dry in an N₂ environment.

Spray processors typically use fresh chemicals for every batch, thus providing pure chemicals for cleaning. Industry use of batch spray processors has declined in recent years, mainly because of the difficulty of achieving very low defect densities while spinning batches of 300-mm wafers. Particle generation can occur from nozzles, seals, etc., and watermarks can form because of inadequate drying of the wafer, as discussed in Section 4.6.3.1. However, owing to batch spray processing's advantages of relatively low cost and small footprint, these systems continue to be used in many nonleading edge applications where defect levels are not as strict. Photoresist stripping, metal stripping, and postcleaning processes are examples of applications that can be effectively accomplished using batch spray.

4.7.3 Single-Wafer Chemical Processors

When single-wafer cleaning equipment was initially introduced, the target processes were specialized steps such as backside and bevel cleaning. However, starting with BEOL cleaning processes in the late 1990s, use of single-wafer systems has expanded such that now they are used in many applications across leading edge fabs, and now the market for single-wafer equipment well exceeds that of batch tools. Especially in logic manufacturing, critical applications such as FEOL precleaning and postetch/ash cleaning processes are typically carried out on single-wafer machines. Single-wafer processes have been heavily adopted based on their uniformity and process control because of the individual processing of each wafer. Typically, the wafer sits horizontally on a chuck and chemical is dispensed from a nozzle above the spinning wafer. The nozzle is frequently attached to an arm that can scan across the wafer at variable speeds during the process. Advanced systems additionally have nozzles that can dispense chemical to

the wafer backside. In addition to double-sided chemical dispense, many single-wafer processing modules have side-specific cleaning, the ability to isolate and clean only one side of the wafer or in some cases to process the two sides with different chemical solutions. Cross-contamination that occurs in batch immersion systems, such as particles migrating from the back surface of a wafer to the front surface of the adjacent wafer, or from edge onto front surface, can be avoided by the use of single-wafer spin tools.

The materials used for single-wafer processing equipment are similar to those required for batch spin-spray systems. Multiple chemical solutions can be sequentially used and the dryer can be integrated in the same module. For physically enhanced particle removal, various jet spray nozzles are available. IPA drying, including the STG type [148], is available for watermark-free drying. For single-wafer tools, to be able to approach throughputs of batch spin-spray or immersion systems multiple processing modules are placed on one platform, resulting in a tool that may contain from 4 to 20 modules. Each module must perform repeatably with respect to the other modules on the platform (chamber matching).

4.7.4 Rinsing Processors

Many types of rinsing and drying equipment exist for IC manufacturing. The rinsing equipment can be divided into three types:

1. Dump rinsing (wet benches)
2. Overflow rinsing (wet benches)
3. Spin-spray rinsing (batch spray tools, single-wafer tools, and occasionally wet benches)

Typically, the rinse process uses ambient-temperature low-resistivity DIW of about 13–18 M Ω -cm. The wafers are rinsed until the outgoing H₂O from the tank or the spin-spray station is at the same resistivity as the incoming H₂O.

4.7.4.1 Dump Rinsing

After steps using concentrated chemicals, a commonly used immersion rinse tank is the quick dump rinser (QDR) with or without a top spray. Wafers are exposed to a spray of ambient-temperature DIW, which fills the rinse tank, or are initially immersed into an already water-filled tank. When the tank is full, the H₂O is dumped from a valve at the bottom. This cycle is repeated several times. The advantage of this configuration is that it removes chemicals from the wafer surface very rapidly, periodically dumping the H₂O, which contains the chemicals. Chemicals such as SPM and H₃PO₄, which are difficult to remove because of their high viscosity, are effectively rinsed using quick dump tanks. This method is very good for rapidly removing the chemical off the

wafer surface and is used in many commercial immersion wet processors. One drawback, however, is that these rinses consume a large amount of water. In addition, spray nozzles tend to generate particles and grow bacteria. Also, QDR tanks generally produce turbulent convective currents in the H_2O , which increase particle mobility to the wafer surface. Hence, the QDR is a poor choice from a particle control point of view, but a very good choice for rapidly removing chemicals. Fig. 4.7-18 shows results from a QDR using numerous rinse cycles after SPM processing to minimize the amount of sulfate residue left on the wafer surface [67].

4.7.4.2 Overflow Rinsing

Overflow rinsing in an immersion wet bench is a common method of performing both the initial rinse after chemical processing and the final rinse before drying. Wafers are exposed to a steady flow of H_2O that rises from the bottom of the rinse tank, then overflows over the top, and finally into the drain. The term, “cascade rinsing” is also used. There are no sprays used. However, the viscous chemicals are not effectively removed from the wafer surface in a cold cascade overflow rinse. Therefore, overflow rinsing is most commonly used after more diluted chemical steps, such as SC-1, SC-2, and DHF. However, this method tends to add fewer particles to the wafer than the QDR.

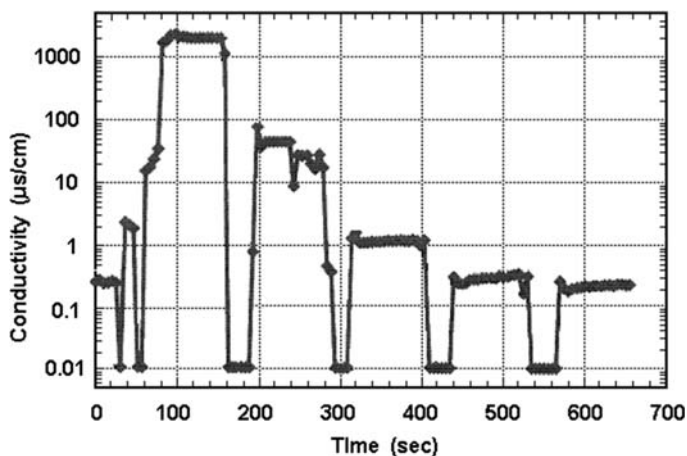


FIGURE 4.7-18 Results from a quick dump rinser using numerous rinse cycles after sulfuric acid/hydrogen peroxide mixture processing. Conductivity decreases after each rinse cycle to a steady-state value after approximately the fifth rinse cycle [67]. Reprinted from R.P. Chiarello, R. Parker, C.R. Helms, W. Chen, S. Tang, L.J. Cook, *Optimization of post sulfuric acid/hydrogen peroxide dump rinsing process*, in: *Symposium Proceedings, Science and Technology for Semiconductor Surface Preparations*, vol. 477, Material Research Society, Pittsburg, PA, 1997, pp. 533. Used with permission.

4.7.4.3 Spin-Spray Rinsing

Spin-spray rinsing can be performed in a centrifugal spin-spray chemical processor after the chemical processing step without moving the wafers to another station. Immediately after the chemical processing is complete, DIW is sprayed onto the wafer to remove the chemicals. The wafers continue to spin during the spray-rinsing process, effectively removing the chemical. Since a spin dryer contains moving parts, spray nozzles, and various gaskets and seals that can be sources of particles, these must be replaced periodically.

Historically, final spin rinsing and drying was often performed after processing in an immersion wet bench. However, this has become rare because of the high occurrence of particles and watermarks. In particular, conducting a centrifugal rinse in the same module as the dryer results in the generation of micrometer- and submicrometer-size H_2O droplets, which may not be effectively removed in the dry cycle. A dry-only mode is by far a lower particle adder than a rinse-and-dry mode. If a spin dryer is used, it is recommended that the rinsing process be accomplished in the rinse tank, and not the spin dryer.

Particle control in the rinse tank becomes especially critical for wet etch processes using HF chemistry. Hydrophobic wafers exiting the HF bath and entering the rinse tank are very susceptible to particle deposition. Menon et al. compared various rinse tank configurations for the effectiveness of cleaning etched hydrophobic wafers [61], as shown in Fig. 4.7-19. Quick dump rinse

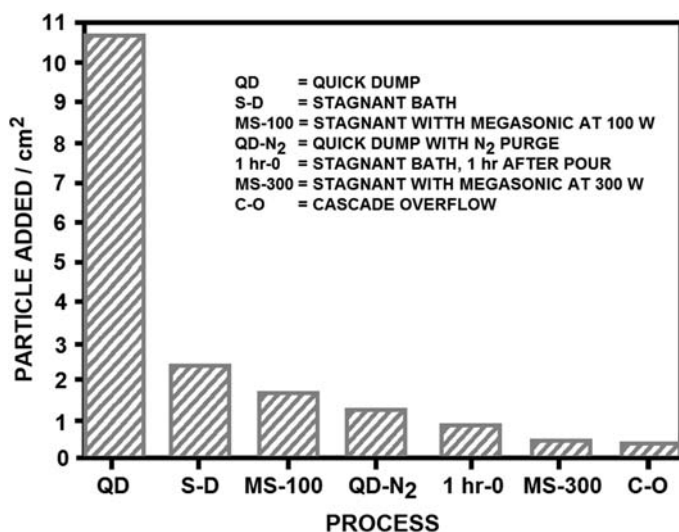


FIGURE 4.7-19 Effects of rinse tank configuration on particle contamination on etched wafers (particle sizes $0.2\text{--}0.5\ \mu m$; initial particle count $<1.2/cm^2$) [61]. Reprinted from *Microcontamination Magazine*, Tom Cheyney, Los Angeles, CA, Copyright 1989. Used with permission.

tanks performed poorly with respect to particle addition during processing, whereas rinse tanks with megasonic transducers performed well. Major sources of particulate contamination during wet-chemical cleaning are often in the rinse tank and spin dryer, and not in the chemical bath. Careful selection of rinse tank and dryer configuration is therefore critical.

4.7.5 Advantages and Disadvantages of Cleaning Equipment

Compared with batch immersion systems, single-wafer cleaning systems have advantages with respect to wafer process control and defect minimization. Single-wafer processors have minimal shielding from chemical exposure as compared with batch processing tools. Multiple touch points, closely spaced wafers, and the fixtures used to hold the wafers can have a detrimental impact on the process uniformity across the wafer and across the multiple wafers in a batch process. In a single-wafer tool, chemicals can be heated and blended very close to the dispense point, and thus, flow through the nozzle can be controlled to optimize the uniformity across a wafer and from run to run. With optimization of various process parameters, both single-wafer systems and batch systems can achieve very good uniformity. Care must be taken to avoid contamination between batch immersion tanks; transfer of contaminants from one wafer to another in the same batch, which can occur in wet bench tanks, is avoided when using single-wafer systems. Complete rinsing of the chemical between batch tank processing is essential. For single-wafer systems, it is important to rinse the entire chamber after each chemical step when multiple chemicals are processed in the same chamber, although the concentric chambers allow chemical exhaust and chemical drains through separate ducts and lines. Both types of equipment are capable of single pass or recirculated (multiuse) chemical processing, although the former tends to be preferred for all but the most expensive chemistries because of demands for high purity and low defects.

The use of spin processing in single-wafer tools enables high flow velocity close to the wafer, which allows faster diffusion of the impurities away from the surface of the wafer [149]. This has proved to be particularly effective for polymer removal processes.

One potential disadvantage of single-wafer spin systems is electrostatic charging of the wafer, particularly during the DIW rinse of the spinning wafer [55]. This has been shown to affect device properties such as GOI; discharge may cause arcing during the process and physical pattern damage can occur. Reducing the resistivity of the rinse water by adding CO₂ has become a common strategy in single-wafer systems, to reduce surface charging to a level similar to that seen in immersion systems [150].

4.7.6 Auxiliary Equipment

4.7.6.1 Chemical Distribution System

Chemical distribution systems consist of pumps, chemical storage tanks, and piping that are required to automatically dispense chemicals to the process equipment. These systems improve safety and reliability, eliminate the need to bring chemical containers into the cleanroom, incorporate multiple prefiltering stations to reduce particles, and reduce the variability associated with manual operation [151]. The chemical compatibility and cleanliness of storage drum and piping materials, the filtration scheme, and the particle generation characteristics of pumps and valves are important aspects to provide low contamination levels at the point of use.

Current IC production requires chemicals with particle levels of less than ~ 11 particles/mL of size greater than $0.02\ \mu\text{m}$ according to the International Technology Roadmap for Semiconductors (ITRS) [152]. Although chemical suppliers are working toward such levels of particulate contaminants in incoming chemicals, the purity at the point of use depends entirely on the filtration scheme in the chemical distribution system and at the recirculating process tank. Fig. 4.7-20 depicts a schematic diagram of a typical bulk chemical distribution system [52], where the actual systems are compact and are remotely located from the process equipment. The chemical drums, storage tanks, pumps, and filter banks are usually located in an area isolated from the main cleanroom. Chemicals are pumped from the source drum into storage tanks, often called “day” tanks, which may or may not have a N_2 blanket above the liquid surface. From the storage tanks, the chemical is continuously recirculated through a bank of filters up to the point of connection to the process tools. Fluid flow within the distribution system is accomplished either with a pump or pressurized N_2 gas. On demand, the chemical solution is drawn from the central recirculation loop to fill a process tank. When the process tools do not need fresh chemicals, the chemicals are continuously circulated

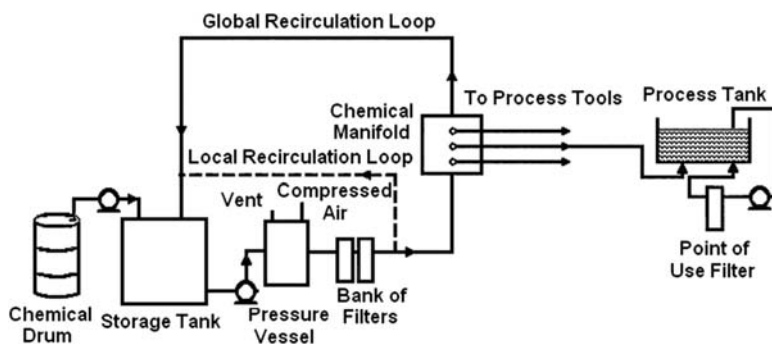


FIGURE 4.7-20 Schematic diagram of a typical bulk chemical distribution system [52]. Reprinted from *Microcontamination Magazine*, Tom Cheyney, Los Angeles, CA, Copyright 1990. Used with permission.

through the filters and the storage tank. Some facilities prefer to have a local recirculation loop that keeps chemicals moving through the storage tank and the filter bank. Automatic systems for spiking of chemical to replenish the bath while wafers are being processed are commercially available. Depletion of the H_2O_2 , especially in SPM baths and also in SC-1 and SC-2 baths, can be remedied by continuous replenishment of chemicals to prevent diminishing of the cleaning ability.

Filters capable of efficiently removing particles $<0.1\ \mu\text{m}$ are commercially available. However, there are compatibility issues that must be carefully examined before selecting a filter for a specific process. Fluorocarbon resins, such as PFA (polyfluoroalkoxy), PVDF, and PTFE, are generally compatible with most inorganic acids and alkalis [153], according to a compatibility chart of filter materials with commonly used chemicals [154]. Often, the membrane of a filter is compatible with the chemical being filtered, but the cartridge material, filter housing, or the O-ring is not. This can result in a change in material composition, leading to gas permeation, polymer degradation, particle shedding, and overall loss of chemical quality. Gotlinsky [155] warns that the material compatibility issues sometimes leads to the selection of Teflon as the material of choice for all filters, whereas lower cost alternatives may be available. Hence, a systematic program for selection, maintenance, and replacement of filters is essential to successfully control particles in a wet-chemical process tank.

For H_2SO_4 , H_2O_2 , NH_4OH , and HCl , 6–10 min of operation in a recycle mode is sufficient to bring the counts well below those observed in bottled chemicals. The particle concentrations at $0.3\ \mu\text{m}$ are typically two to five times that at $0.5\ \mu\text{m}$. Gruver et al. [156] report the performance of another chemical distribution system for all key chemicals; the final particle count at the point of use should be below 0.50 particles/mL. On an average, the chemical distribution system with its filters can provide a 100- to 500-fold decrease in particle levels. This reduction does not include any point-of-use filtration. Rosenfeld et al. [157] published results of another distribution system. Their results indicate a 35- to 300-fold decrease in particle levels from the drum to the point of use, depending on the chemical composition and type of filters being used.

When designing a chemical distribution system, it is worthwhile to include sample ports at various points in the system to extract chemicals that can be analyzed for particles. Sample ports are recommended after the drum, the pumping unit, and the storage tank, and at the point of connection to the process equipment. Also, building in some redundancy in pumping capability is useful as chemical usage demands increase or if one of the pumps is being repaired.

4.7.6.2 Point-of-Use Filtration

State-of-the-art fabs generally prefer to do all their DIW filtration at a central purification facility, and not at the point of use. Point-of-use H_2O filters are breeding grounds for bacteria because of the low volumetric flow rate of H_2O

through each filter. Also, since most filters are not compatible with O_3 , they cannot be used with DIW/ O_3 .

Wafer cleaning processes that are conducted in an immersion tank or spray processor accommodate point-of-use recirculation filters. Recirculation tanks, with continuous filtration using advanced polymeric material, are available for processes that are conducted at operating conditions ranging from room temperature to as high as 180°C . Some suppliers offer recirculation systems for SPM baths that are maintained at $120\text{--}130^\circ\text{C}$ and for 85 wt% H_3PO_4 maintained at 180°C . Systems for heating solvents and semiaqueous solvents are also commercially available.

As with filters for chemical distribution systems, it is important to select a point-of-use filter that is compatible with the chemical being filtered. Compatibility charts are available from most filter manufacturers. Unlike DIW filters, bacterial growth has not been a problem with chemical filters. Before a filter is installed, it is a good idea to consider the following:

1. Is the filter chemically compatible?
2. Does the filter need prewetting before it can be exposed to the chemical?
3. What are the ionic components that are extractable from the filter, and will they detrimentally impact the wafers?
4. Will the filter handle the viscosity of the chemical without clogging and creating a high-pressure drop, which may ruin the pumps?
5. How often does the filter have to be replaced?

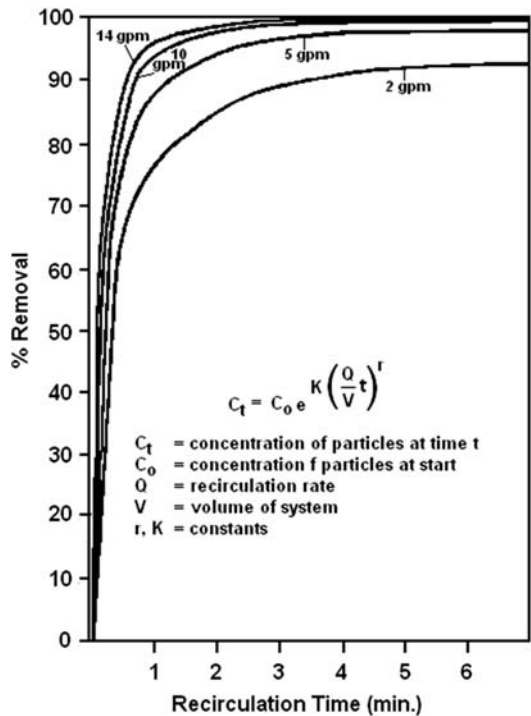
Once a point-of-use filter is installed, it will shed particles for a short period of time. Also, particle levels in a chemical do not decrease instantaneously after a recirculation pump has started to pass the chemicals through the filter. Fig. 4.7-21 illustrates the effects of various flow rates on the removal of particles from recirculating acid-etching immersion tanks [155]. As the recirculation flow rate increases, the PRE also increases. This figure illustrates that the particle level can drop from 100 to 0.40 particles/mL in 4 min at a recirculation rate of 53 L/min. For viscous materials, including some buffered HF solutions [45], it may take more than 1 h for the particle level to attain its lowest steady-state value. When chemicals are supplied manually in bottles to such a recirculating tank, it is necessary to provide adequate time for particle levels to stabilize.

Methods for controlling particles in chemicals by filtration and chemical recirculation have greatly improved year after year. Particle levels in incoming liquids and at the point of use show that a 500-fold decrease in particle levels can be achieved through optimization of chemical delivery systems.

4.8 CURRENT AND FUTURE CHALLENGES

In recent years the pace of change in semiconductor manufacturing technology has accelerated dramatically. The historical model in which continued

FIGURE 4.7-21 Particle removal versus recirculation time in an acid batch immersion tank [155]. Used with permission from *Microelectronics Manufacturing and Testing*.



lithographic scaling defined the industry’s remarkable productivity growth has given way to a new era in which increasingly radical changes in materials and device structures are required to maintain the pace. Jettisoning materials that have serviced the industry for decades in favor of as-yet little characterized new materials is now seen not merely as a source of research topics but a necessity for production of new device generations. The ITRS [152] is a useful guide to these changes.

Wafer cleaning has always needed to advance to meet the requirements of scaling. Demand for fewer particles, metals, and organic contaminants has historically driven requirements for cleaner chemicals, water, and gases, plus stricter attention to equipment design and materials, as well as decreased human proximity to the wafers in the IC manufacturing areas. However, while those demands still persist, the more pressing challenges result from new materials and device structures. Going forward, the wafer cleaning processes need to be more effective in removing contaminants to lower levels and these processes also must be more selective and provide less etching or roughening of the surface without any harmful effects to the structures and the final device

performance. Accordingly, wafer cleaning issues can no longer be viewed in isolation, but need to be considered as part of the overall process integration.

4.8.1 Front-End-of-Line

4.8.1.1 Logic Technology

In general, the most urgent requirements in the logic technology roadmap depend on the type of product. For high-end microprocessors, transistor speed (i.e., drive current) is of utmost importance. For low-power devices used in mobile phones and other portable electronic products, minimizing leakage current is imperative. In general, the requirements for greater speed and lower leakage must be pursued simultaneously. The latter requirement has led to high-dielectric-constant (high-k) materials replacing SiO_2 or SiON as the gate dielectric, since further scaling of those traditional materials would lead to intolerable leakage by direct tunneling of electrons [158]. Such materials, typically Hf based, have become mainstream in recent logic nodes, including both high-performance microprocessors as well as low-power devices.

In conjunction with high-k dielectrics, the traditional polySi gate electrode has been replaced with metal electrode materials, to match the work function of the high-k dielectric. Various metals have been used by different IC makers, with TiN, Ta, and variants of Al being common. The introduction of high-k and metal gate devices was pioneered into production by Intel for the 45-nm technology node [159].

Challenges for aqueous cleaning processes related to high-k and metal gate stacks include the following:

- Control of the chemical oxide thickness formed by the aqueous surface preparation step before dielectric formation, since the electrical thickness targets for the gate dielectric have become small enough to make the chemical oxides formed by traditional RCA cleaning processes a significant contributor.
- Removal of the remaining high-k dielectric films after etching the gate stack. For SiO_2 gates a simple DHF step could be used. Some of the proposed high-k dielectrics, however, are difficult to etch with sufficient selectivity to SiO_2 (so as to avoid excessive consumption of the isolation regions). HfO_2 represents a particularly difficult case [158,160].
- Removal of high-k dielectric films from the back surface of the wafers after deposition, since the remaining material on the back surface can cross-contaminate other FEOL process systems with metal contamination.
- Particle and residue removing chemistries that are compatible with the chosen metal gate material(s).

Particularly for etching and cleaning with metal gates, new chemistries (including nonaqueous) or modifications to traditional chemistries may be needed.

The most urgent challenge currently facing aqueous postprocess cleaning steps in FEOL logic is cleaning with minimal Si/SiO₂ consumption and without damage to sensitive patterns [152]. Until recently, gate patterns were considered the most sensitive feature with respect to damage for logic devices. However, with the implementation of finFET (fin field effect transistor) architectures since the 22-nm node, these fin structures are extremely sensitive. In the case of fins, the clean immediately following the dry etch that forms the fin is extremely sensitive to damage, because of its thin width and high aspect ratio. Etch residues must be removed without physical damage and the closely spaced features dried without stiction-related collapse. In the case of gate feature patterns, photoresist striping is especially sensitive because of the potential loss of material. After ion implantation with a photoresist mask, a sequence of plasma strip—SPM—SC-1 is typically used to remove the photoresist. This combination leads to:

1. Consumption of Si because of the combination of oxidation and etching.
2. Potential damage to narrow (<2 nm) polySi gate lines from the physical energy used for particle removal.

Silicon consumption, historically, was not a problem, but the number of implantation steps after gate formation has increased as the doping profiles have become shallower; therefore, more Si undesirably is consumed. Of course, SC-1 can be processed at reduced concentration, temperature, and/or time to minimize the consumption, but this can compromise PRE, which also is affected by reducing or eliminating the use of megasonic processing because of pattern damage. Optimization and the development of nondamaging cleaning with megasonic energy or jet spray are being carried out by various research groups.

Logic FEOL integration schemes have demonstrated the use of Ge as a channel material for PMOS (p-channel MOS), because of its high carrier mobility [161]. Currently, SiGe is in production use, with the content of Ge in these films expected to increase until the ultimate limit of pure Ge. Meanwhile, III-V materials have been proposed to realize electron mobility improvement in NMOS (n-channel MOS) [162]. These new materials represent a fundamental shift from the Si-based devices that have dominated the industry for decades. Si readily forms a stable and passivating oxide with a low defect interface; the new materials do not share this property. Materials such as Ge and III-V materials such as GaAs, and InGaAs form oxides that are soluble in water [163]. Therefore, oxidizing solutions such as traditional RCA cleaning processes have much higher etching rates for these materials than they do for Si, and care is required in choosing a cleaning chemistry. For example, Sioncke et al. [163] reported that the traditional 1:1:5 SC-1 solution at 75–80°C etches Si at about 0.82 nm/min, whereas it etches Ge at about 275.5 nm/min and GaAs at 1890 nm/min. They proposed a 1:1:5000 SC1 solution for particle removal, but this presents challenges for process control in

a production environment. DeSalvo et al. [164] proposed a two-step process using H_2O_2 followed by an acidic solution. Such a “digital etch” can remove a controlled amount of GaAs (15 Å in their study) for each application of the two-step sequence. For InP, acidic H_2O_2 solutions with well-controlled mix ratios were proposed by van Dorp et al. [165] to enable a small amount of surface etching (e.g., for particle removal) without roughening the InP surface.

4.8.1.2 Memory Technology

DRAM technology currently uses stacked capacitors with increasingly high aspect ratios that present challenges for cleaning and surface preparation. Stacked capacitors in the presence of liquids can collapse because of the surface tension forces, thus making rinsing and drying difficult. In addition, shallow trench isolation (STI) structures feature high aspect ratios and can suffer from stiction while undergoing drying. As these aspect ratios become more severe, even heated IPA and surface modification drying methods are reaching their limit and supercritical CO_2 -based drying is under consideration, as discussed in Section 4.6.3.2 [145]. Cleaning with selectivity to W metal gates is another challenge in advanced memory.

Flash memory technology, such as NAND, has seen rapid growth in recent years and now represents a significant portion of the overall fab capacity. Scaling of critical dimensions in flash technology has become more aggressive than even DRAM, requiring very strict defect densities. Therefore, flash factories will rely more and more on state-of-the-art cleaning technology. Floating gate NAND has high aspect ratio structures for the STI and gate features that are vulnerable to stiction-driven collapse. Currently state-of-the-art flash memory is in transition toward 3D architectures (V-NAND) that are fundamentally different from historical planar device structures. Although horizontal critical dimensions actually will increase, the multilayered vertical stacks being formed will lead to extremely high aspect ratio structures, in particular holes and slits, that must be penetrated uniformly from top-to-bottom by the wet chemicals used for cleaning and etching.

4.8.2 Back End of Line

4.8.2.1 New Materials

Introduction of novel materials in Cu interconnects integration schemes provides additional challenges for wet cleaning suppliers. For example, new diffusion barrier materials have been recently introduced. For technology nodes above 14 nm, a tantalum/tantalum nitride (Ta/TaN) bilayer is typically used for Cu interconnects. Tantalum provides an adhesion layer between Cu and TaN, whereas the TaN acts as a barrier layer to prevent Cu diffusion into the low-k material [166]. As technology nodes continue to advance and device dimensions continue to decrease, the barrier and liner thicknesses must also be

reduced to minimize overall wire resistance [167]. To precisely control the thickness, the composition and uniformity of the barrier and the liner become critical. IC manufacturers have replaced physical vapor deposition (PVD) Ta with chemical vapor deposition (CVD) Co (cobalt) liners for advanced nodes [168]. The challenge with Co liner is the galvanic corrosion of the Co that occurs when Cu and Co are in contact with one another, which establishes electrical contact in the cleaning solution [169]. Another widely studied liner candidate is ruthenium (Ru) [170]. When Cu and Ru are in electrical contact in the cleaning solution, it is the Cu that is corroded [171]. Wet cleaning formulation manufacturers have put considerable effort into developing cleaning solutions to overcome this issue. In addition to these new liner materials, CVD TiN has been considered as a candidate to replace the PVD TaN barrier layer for the metal 1 contact layer. The challenge, in this case, is to have a cleaning solution that removes the PVD TiN mask (about 300 Å) and still is compatible with CVD TiN barrier.

Reducing the etch stop layer thickness is necessary to realize a lower capacitance for BEOL dual Damascene dielectric layers [172]. Compared with conventional SiCN etch stop layers, AlN (aluminum nitride), Al_xO_y (aluminum oxide), and other Al compounds are attractive because they exhibit a low etching rate with fluorine-containing gas plasma chemistry because of the predominant formation of AlF_3 , a low-volatility by-product [173,174]. Some IC manufacturers have replaced SiCN with AlN or Al_xO_y to improve etch selectivity between the low-k and etch-stop layer. There is a problem when using AlN or Al_xO_y for etch-stop layers in patterned wafers. During low-k etching, the AlN/ Al_xO_y becomes damaged by exposure to fluorine-containing gas plasma and forms AlF_3 , which is nonvolatile, but water soluble. This poses a serious formulation challenge for wet cleaning manufacturers, because the AlN is removed when it is dissolved in DIW during the wet cleaning step [175]. For advanced technology nodes, the cleaning solutions generally remove the TiN hardmask, the AlN etch-stop layer, and the PERs in one single step.

IC manufacturers are considering the use of Co to replace tungsten for the contact plug, barriers, and interconnect lines to reduce contact and line resistance [176]. Cobalt is being considered because it has lower resistivity than some of the barrier materials such as TiN, which can increase the contact volume by using thinner barrier metal, allowing more fill metal, and it has better thermal stability for annealing [177]. Integration will create a challenge for cleaning, in that Co is not compatible with fluoride chemistry [178]. The formulation suppliers are challenged to create a cleaning solution that is compatible with the Co plug and yet capable of removing the TiN mask in situ.

4.8.2.2 Copper Dual Damascene Interconnects

Copper dual Damascene wiring, with lower resistivity than aluminum, is now fully mainstream and will remain the interconnect metal for foreseeable

technology nodes. One method for controlling the cleaning process with the Cu dual Damascene devices is to move from wet benches to spray tools or single-wafer cleaning systems. Single-wafer cleaning has been gaining acceptance in the back-end-of-line process especially for 300-mm wafers at nodes less than 130 nm. Batch processing in wet benches with the traditional and semiaqueous fluoride chemistries can be difficult because of the possible attack of the metal during post-metal cleaning. Rinsing efficiency in batch immersion processing is limited by diffusion, such that the residence time of fluoride at the wafer surface is difficult to control in comparison with faster rinsing single-wafer spin processes. Thus corrosion can be more difficult to avoid in batch immersion systems.

The challenge for any new chemical formulation for <130-nm technology is to be able to affectively clean structures in a very short period and then quickly quench the chemical reaction when the cleaning process is complete. Many IC cleaning recipes for single-wafer spin-spray tools call for a maximum of 2-min cleaning cycles, including the H₂O rinse and drying steps. This is one of the reasons that aggressive formulations have been developed for the single-wafer technology, to enable fast process times and high throughputs with excellent cleaning results.

4.8.3 Manufacturing Concerns

4.8.3.1 Cost

The manufacture of semiconductor chips is expensive. Any company undertaking IC manufacturing must first rigorously measure, and then constantly seek to minimize, its manufacturing costs. Thus the engineer finds himself or herself spending a lot of time and effort on cost issues in addition to the technical challenges of process performance and yield.

It has become common to predict and track costs in detail on a unit process level. The Cost of Ownership model, first developed for the industry by SEMATECH [179], is a useful tool when comparisons of different process options are needed, since the sources of cost and the trade-offs between various options can be complex. The most common decision required is choosing one cleaning tool over another. The cost per wafer pass through a tool is a function of fixed cost (such as purchase price of the tool), variable cost (such as consumables, chemicals, and labor), yield loss, throughput (wafers/h) capability, and the percent of time the tool is actually processing wafers (a function of the tool's reliability). Thus a trade-off might be, for example, between one tool with higher throughput and another with lower consumables (e.g., chemicals, water, and gases) cost plus a lower price. Cost modeling provides a quantitative comparison [179]. Semiconductor chip makers employ their own various models.

4.8.3.2 *Environment, Safety, and Health*

In any industrial environment, strict attention must be paid to issues of ESH. Semiconductor manufacturing systems employ hazardous chemicals, high levels of electrical energy, and automation that is capable of causing serious injury.

In the area of wet cleaning, chemical and water consumption are the ESH areas that have historically received the most attention. Efforts have been made to reduce concentrations and volumes of chemicals used, which has the combined benefit of reducing risk of exposure to personnel and effects of disposal. Of course, cost benefits can simultaneously be realized. The industry sets ESH goals and guidelines that are outlined in the ITRS [152]. To date, H₂O use reduction through rinse optimization and energy use reduction through exhaust optimization have been among the focal points.

III-V materials pose a risk for evolution of arsine or other harmful gases when the material is etched. This will need to be well managed with the use of proper abatement equipment and techniques.

4.9 SUMMARY

For many years there have been predictions of aqueous cleaning being replaced by gas phase or other dry cleaning methods. Although the application of dry surface preparation and etching steps has undoubtedly increased, the number of wet processing steps has continued to increase and in particular aqueous processes are even more common since very dilute H₂O-based chemistries present both ESH and cost benefits. Also, it is generally true in semiconductor manufacturing that existing methods will be extended as long as possible. As has been discussed in this chapter, aqueous cleaning offers unique advantages for removing a wide range of particulate, metallic, and organic contaminants in a robust manner that can still be selective to critical materials on the wafer.

Cleaning processes were once regarded to be relatively simple in comparison with other areas of manufacturing. And although lithography, for example, understandably continues to garner more attention, cleaning is recognized more and more as a critical process category that needs to be well characterized and engineered for a technology to succeed. In the recent few decades, researchers in industry, universities, and laboratories have worked to elucidate the mechanisms of aqueous and other cleaning processes, producing a tremendous number of publications and improved understanding. Furthermore, attention is given to cleaning at earlier stages in the technology development cycle, especially in light of the new materials being introduced on wafers. Cleaning is being viewed more in an integration context as opposed to a set of isolated processes, especially with regard to removing contaminants

selective to new materials and structures on the wafer. This trend can be expected to continue.

Finally, environmental, health, and safety issues will remain of great concern. These also need to be addressed as early as possible in the development cycle, although in the past this has not always occurred. Although there is a common perception that addressing ESH issues necessitates compromising cost and process results, it has been demonstrated that processes engineered from the beginning with ESH in mind can also provide the best performance at the lowest cost.

ACKNOWLEDGMENTS

Excerpts from the first and second editions are used and are credited to Venu B. Menon, Robert P. Donovan, Don C. Burkman, Donald Deal, Donald C. Grant, Charlie A. Peterson, and Robert J. Small.

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